ENT

Workbook

Dr. Shubham Mittal

MBBS, DNB-ENT, MNAMS

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

© All Rights Reserved with MIST. No part of this publication shall be reproduced, copied, transmitted, adapted, modified or stored in any form or any means. If found at any point of time, shall be penalized as per the Courts of Law.

Contents

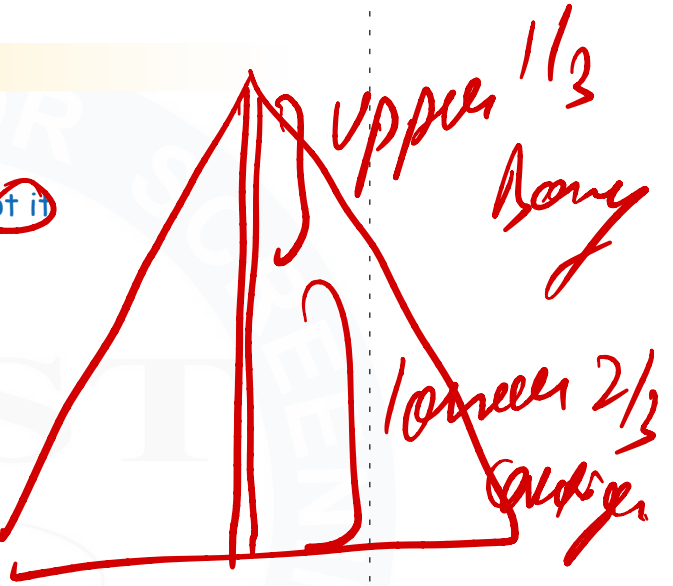
1. NOSE	05 - 40
2. EAR	41 - 101
3.LARYNX	102 - 114

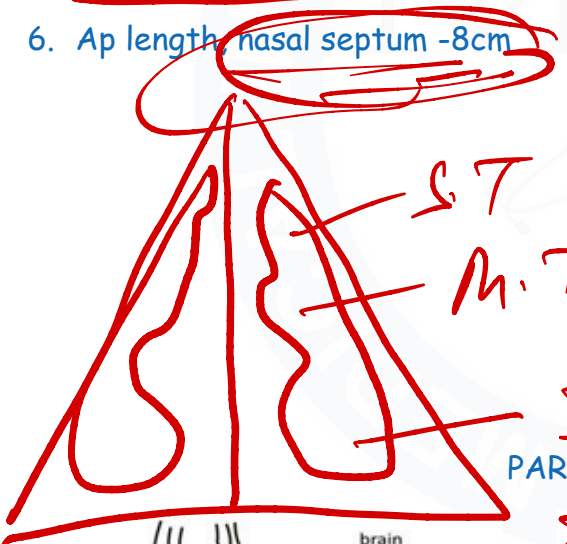
MEDICAL INSTITUTE FOR SCREENING TEST
MIST

ENT

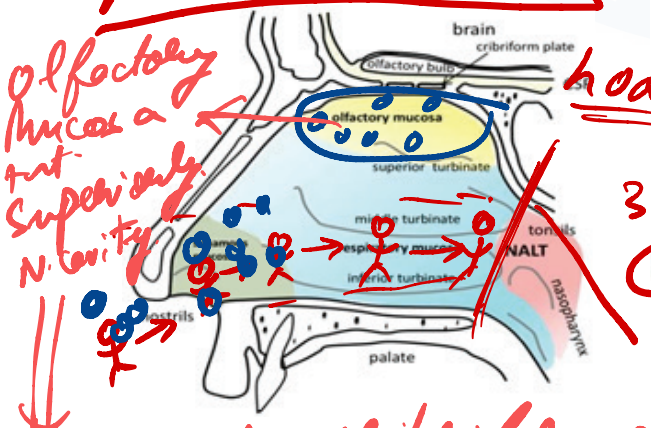
ANATOMY OF NOSE

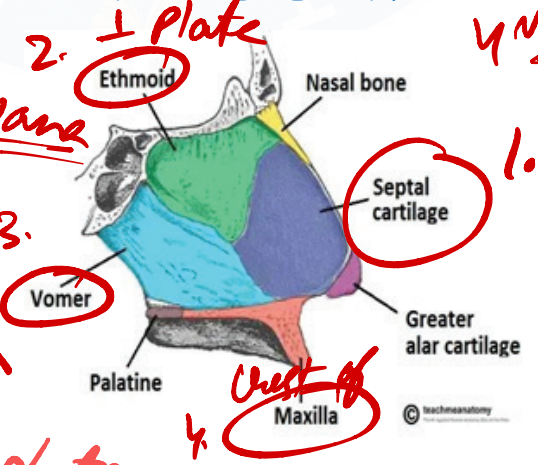
1. Upper 1/3rd bony, lower 2/3 cartilagenous
2. All turbinates are part of ethmoid bone except it
3. Largest -it
4. Ph of nose 7
5. Choana is a junction nose and nasopharynx
6. Ap length nasal septum -8cm





PARTS OF NASAL SEPTUM





Olfactory mucosa at superiorly. N. cavity

choana

2. ⊥ Plate

4 major parts

1.

3.

4.

Close to the cribriform plate.

Active space

MINOR PARTS OF SEPTUM

- Spine of frontal bone
- Spine of maxilla
- Palatine bone
- Rostrum of sphenoid

PARA NASAL SINUS

Function
- drainage of secretions.
- makes skull lighter.

Air filled space around nasal cavity.

SWELLINGS

Maxillary
Ethmoid → Ant.
→ Post.
Frontal
Sphenoid

MODIFICATION OF PNS CELLS

Largest sinus - 15 ml ⇒ Maxillary.

Most functional pns - 15 air cell ⇒ Ethmoid.

Agger nasi → Ant. most ethmoidal air cell.

Bulla ethmoidalis → largest ethmoidal air cell.

Roof of maxillary sinus - floor of ant. Ethmoid+orbital floor ; haller cell

Most posterior - onodi - on optic nerve.

proximity to

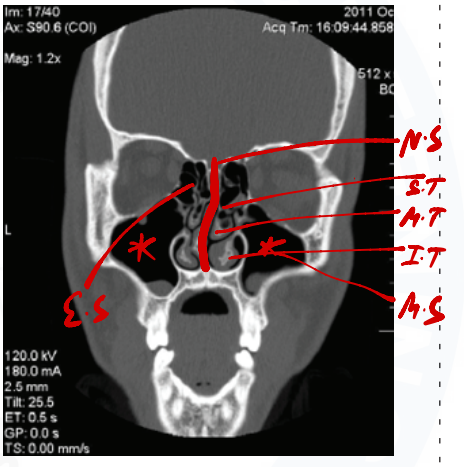
MIST

CT FINDINGS

Air - black White -bone

Between soft tissue Ct - bone

Mri - soft tissue

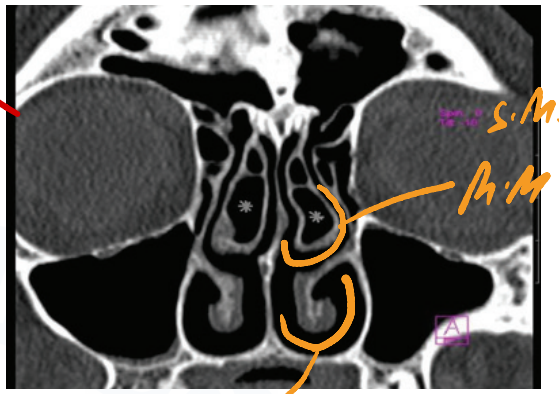


Active space

CHONCHA BULLOSA

Pathological

Air inside Middle Turbin



Glabella- most prominent part frontal bone

Radix-upper edge nose

Nasion- naso frontal angle-134 degree

1st Br. of Facial Ne.

Greater Superficial petrosal nerve.

Crocodile tears → GSPN

MEATUS

Inf meatus → Nasolacrimal duct [Hasner's Valve]

Middle meatus → Maxillary, Frontal, Ant. ethmoid

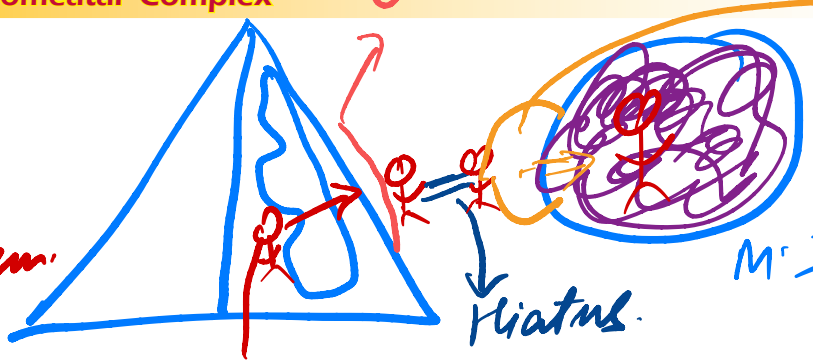
Superior meatus → Post ethmoid.

Sphenoethmpoidal recess → Sphenoid.

Osteomeatal Complex

Uncinate Process

Infundibulum



1. Uncinate process.
2. Hiatus.
3. Infundibulum.

Hiatus.

MIST

Active space

FESS → Functional Endoscopic

Sinus Sx.

APPEARANCE OF SINUSES

- Maxillary
- Frontal- does not develop untill 4 yr old
- Sphenoid
- Ethmoid - 1year
- Last sinus to develop is frontal
- May grow upto 14 years

1st PNS → Maxillary

Birth → M + Ethmoid.

SINUSITIS ⇒ inflammation of PNS

1. Viral -rhino Bacterial-strepto, "Sir" steph. aureus.
2.
3. Fungus-aspergillus fumigatus more in nose

Niger more in ear

→ Adults

Sinus involvement- m > f > e > s

Paeds- e > m > f > s

Children

Ioc- diagnostic endoscopy

Radiological ioc-ct

Toc-fess

DNE

Diagnostic Nasal Endoscopy

Active space

CRITERIA FOR DIAGNOSIS

2 major + 1 minor

Major

Anosmia

Blockage

Congestion

Discharge

Facial pain and fever and halitosis - minor

Active space

SWELLINGS

Cheeck/ dental/lower eyelid

Maxillary

Root/radix/dorsum/upper and lower eyelid

Ethmoid

Office headache/ frontal headache/more in morning less in evening

office hours.

Frontal

Retro-orbital/occipital/vision (No swelling)

Sphenoid

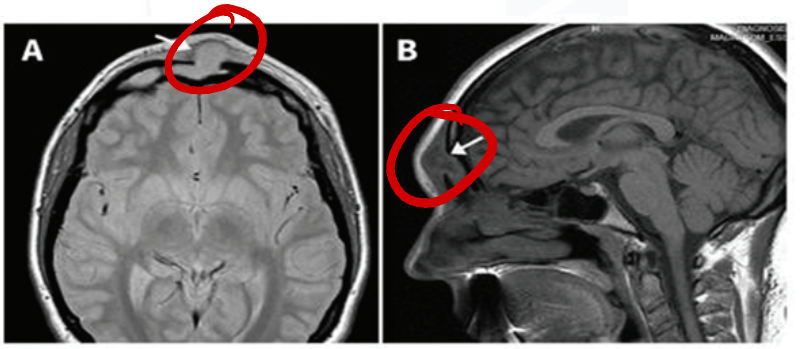
Active space

COMPLICATIONS

1. Frontal sinus: mucus----> pus----->osteomyelitis

A/k/a potts puffy tumour

2. Orbital abscess- ethmoid sinus



Active space

ALLERGIC FUNGAL RHINOSINUSITIS

AFRS

Characteristic question asked

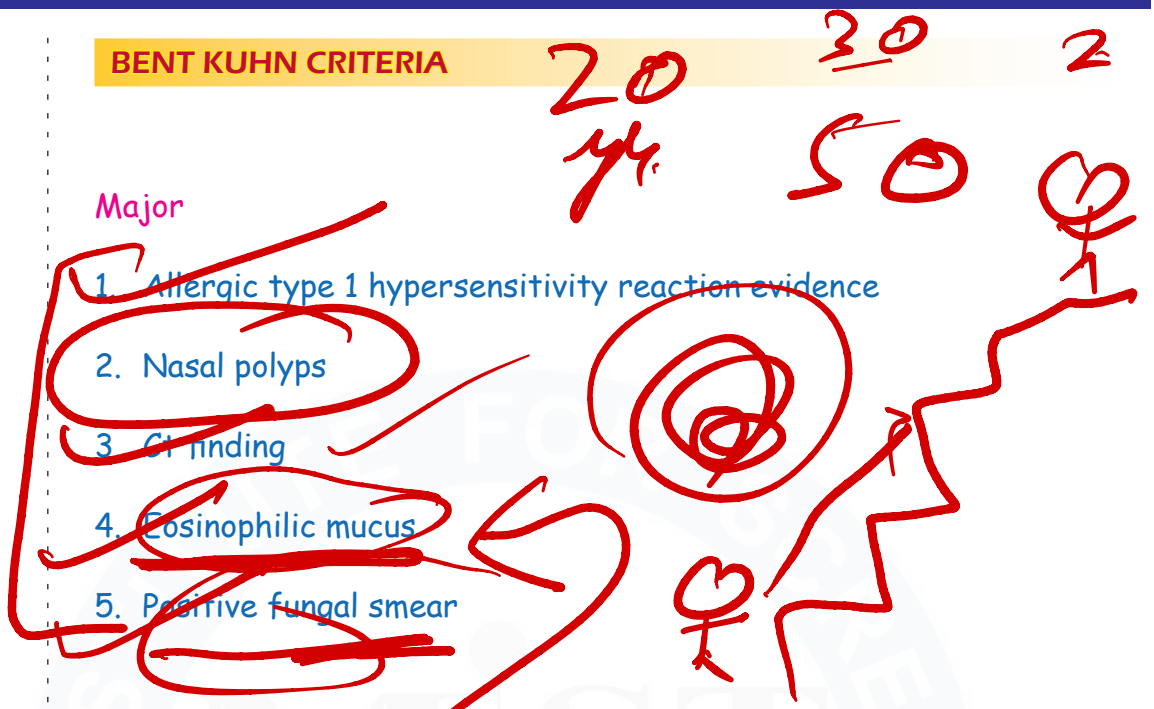
1. Peanut butter appearance of nasal secretion
2. X-ray double density sign

CT

Some areas have more & some less deposition of Calcium & fungus

Active space

BENT KUHN CRITERIA



Major

1. Allergic type 1 hypersensitivity reaction evidence
2. Nasal polyps
3. CT finding
4. Eosinophilic mucus
5. Positive fungal smear

Minor

1. Asthma
2. Unilateral predominance(disease)
3. Bone erosion in radiographs
4. Positive fungal culture
5. Charcot leyden crystals
6. Eosinophilia in serum

A → Asthma
B → Bone erosion
C → charcot leyden
D → Disease (U/L)
E → Eosinophilia
F → Fungal culture

Active space

RHINITIS

Allergic rhinitis

→ Type 1 IgE

Signs → Allergic Salute

→ Allergic Shiners (Blackish discs below eyes)

T/t ⇒ 1. Antihistamines

2. (Xylometazoline / Oxymetazoline) ← α-1 agonist

3. OMALIZUMAB

Anti IgE ab drug.

Hypertrophic rhinitis

- Mulberry like nasal mucosa (hallmark) T/t - ktp laser

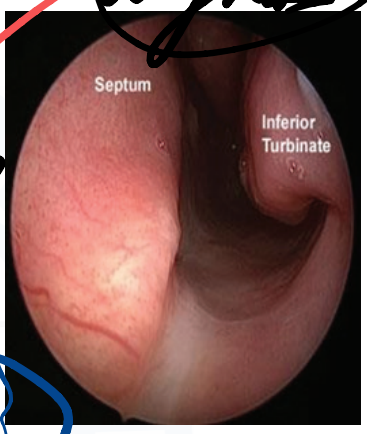
Active space

MIST

ATROPHIC RHINITIS

Hernia

Merciful anosmia



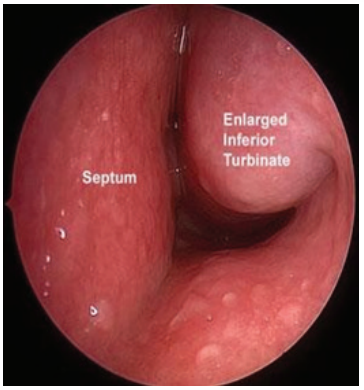
Alkaline nasal douche

Surgery- young operation: block nose for 6 months

Modified young: partial closure

Inj. Teflon to lateral wall

Lautens laser surgery -latest new method



MIST

DRUG INDUCED RHINITIS

Vasodilators like

1. Antihypertensives
2. Ocps

Cause it

T/t - vasoconstrictors- oxymetazoline

RHINITIS MEDICAMENTOSA

VASOMOTOR RHINITIS

Vidian neurectomy done to treat it

Active space

TRAUMA

- If superior/ posterior - skull fracture
- If anterior/inferior/lateral

Multidirectional facial fracture Most common is nasal fracture 2nd m/c is zygomatic fracture

NASAL BONE FRACTURE

Reduce fracture immediately

If oedema present—>antiinflammatory for 5 days—->reduction

Reduction if done in less than 3 weels - do closed reduction

More than 3 weeks - do open reduction

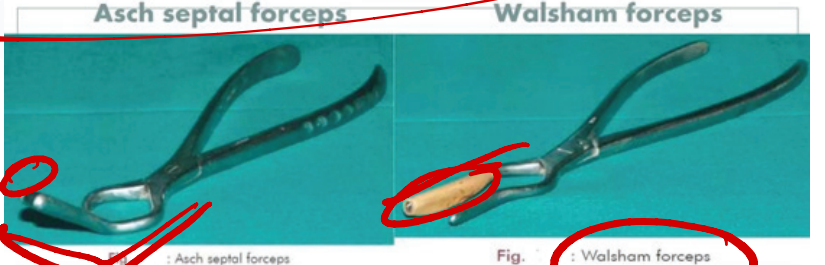


Fig. : Asch septal forceps

Fig. : Walsham forceps

Asch forceps	Walsham forceps
Angled blades	Straight blades
No rubber tube on the tip	Rubber tube on one tip to protect skin
Gap seen between prongs on approximation	No gap seen on approximation
Used to elevate and straighten the septum	Used to refracture and realign nasal bones
Stout	Smaller in size

Active space

ZYGOMATIC FRACTURE

Called as tripod fracture:

1. Frontozygomatic fracture suture
2. Zygomatic arch
3. Zygotemporal fracture suture

Active space

LE FORTE FRACTURE

Facial maxillary fracture

Ist

- low maxillary fracture
  - Floating palate
  - Guiren sign → Bluish purplish discolouration of palate

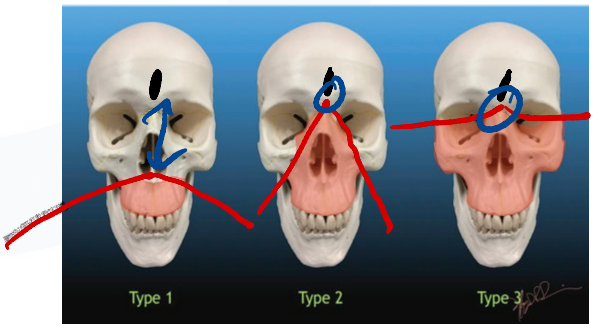
2nd

- Pyramidal fracture

3rd

- Craniofacial fracture

→ C.S.F Rhinorrhea.



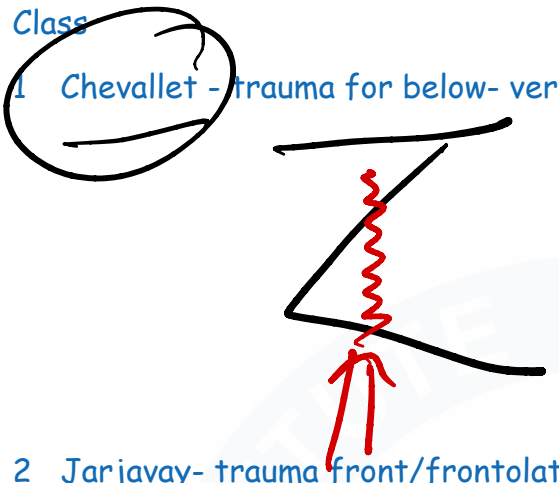
Active space

MIST

DIRECTION OF TRAUMA

Class

1 Chevallet - trauma for below- vertical #



2 Jarjavay- trauma front/frontolateral-horizontal #

3 Multidirectional

Active space

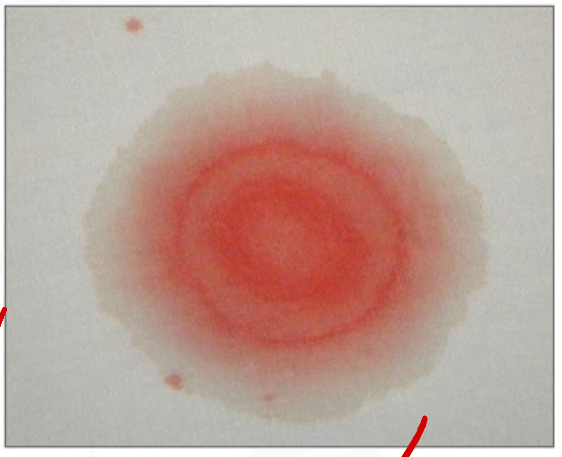
Nose.

CSF RHINORRHEA = discharge.

Trauma to cribriform plate

90% traumatic- m/c surgical trauma

Non traumatic- due to increased i.C.T



Target sign / Double ring / Halo sign.

Single ring sign - non traumatic Double ring/target/halo sign- traumatic

Ix of CSF → Biochemical → β_2 transferrin.

→ Radiological C.T scan. MIST

Active space

T/t ⇒ Conservative.

CSF RHINORHEA VS SINUSITIS

Cause	Trauma.	Infections.
Type of d/s	Watery.	Mucoid.
Sniffing	X	✓
Glucose	✓✓	XX sht.
Beta 2 transferrin test	✓	

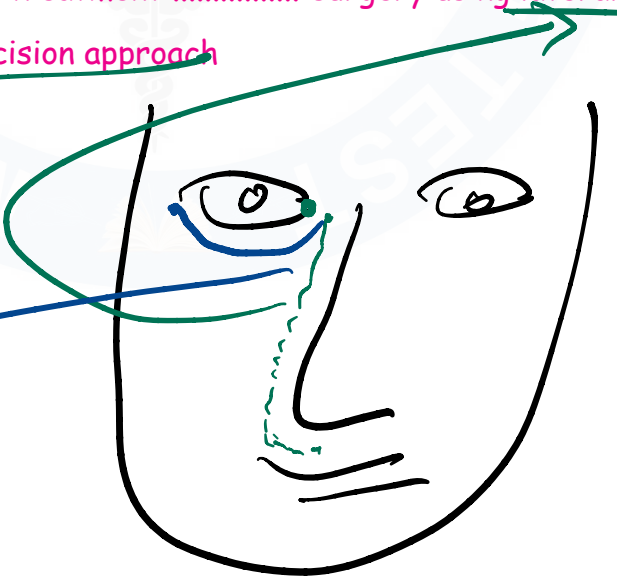
Active space

MALIGNANCY NOSE

- M/c benign tumour - capillary hemangioma (site-little area)
- M/c site cavernous hemangioma -it
- M/c site malignant melanoma- nasal septum
- M/c site papilloma - skin behind columella/vestibule
- M/c malignant -scc —> lateral wall of nose

of Nose

- All tumour treatment .....>surgery using lateral rhinotomy/ moore's incision approach



Weber fergusson incision.

INVERTED PAPILLOMA/RINGERTZ TUMOUR/SCHNIDERIAN PAPILLOMA/TRANSITIONAL CELL PAPILLOMA

Tumour that arises from lateral wall of nose Premalignant tumour

M>f

>40 years

Inverted as per hpe finding

Rather than growing outwards, it grows inwards towards Stroma.

In both scc and inverted papilloma there is invasion of maxillary sinus

But if

Wall of max sinus damage present - scc If not-i p

Active space

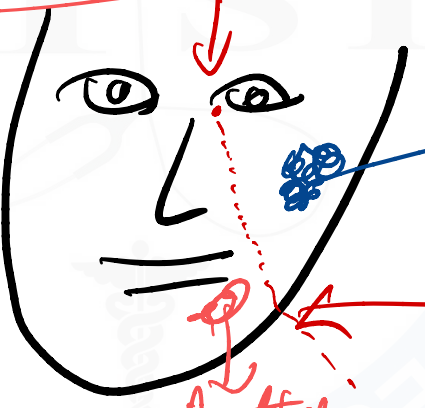
TUMOUR OF PNS

1. M/c benign tumour of pns-osteoma- seen in frontal sinus
2. M/c malignant tumour of pns-scc - in maxillary sinus - people working in nickel industry
3. 2nd m/c is adeno carcinoma - in ethmoid sinus- wood industry

For these t/t is surgery + radiotherapy

MAXILLARY SINUS CA

OGHRENS CLASSIFICATION



GRANULOMATOUS DISEASES OF NOSE

1. M/c site of leprosy and tb - ant.Part of septum and i.T
2. Stage of syphillis involving nose - tertiary M/c symptom - nasal obstruction
3.
4. Lupus vulgaris-m/c site is mucocutanous junction of nasal septum

Delayed Type IV

Active Space

RHINOPHYMA

Potato Nose / Whisbey Nose

Om Puri

Hy pertrophy of Sebacous glands of Nose.

T/t → Laser.

RHINOSCLEROMA

DOC : S + T.

K. rhinoscleromatis

Klebsiella.

Woody nose/tapir nose/hebra nose/tapir nose Stages

Atrophic- carpenter glue like discharge Granulomatous- hebra nose

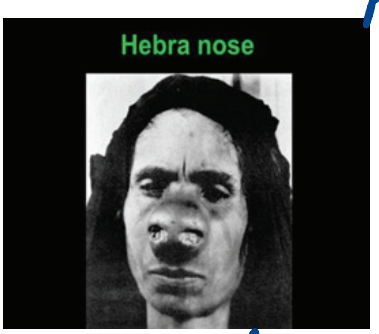
Cicatricial- tapir nose

Rhinoscleroma a/w subglottic stenosis with respiratory failure

Russel Bodies (Eosinophills)

Mickulicz cells [Foam cells]

Pathological cells of Rhin.



Hebra.

Tapir

MIST

Active space

RHINOSPORODIOSIS

South india

Excision and cautery at base

Dapsone

South India.

Salt, sea water.

Rhinosporidium

See beer

aquatic Protozoa.

MULBERRY LIKE POLYP

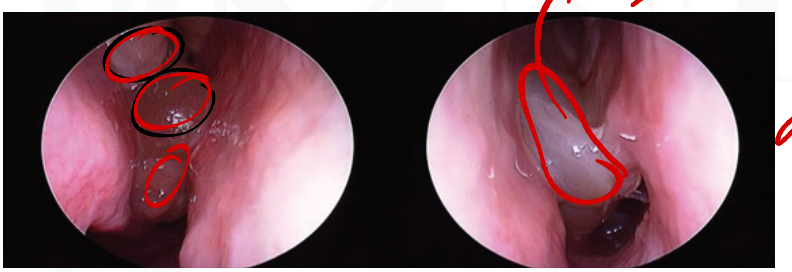
Active space

NASAL POLYP

Bernoulis principle

1. Kartagener syndrome- nasal polyp + bronchiectasis + dextrocardia
2. Young syndrome- np + azoospermia + bronchiectasis
3. Samters triad - np + bronchial asthma + aspirin sensitivity

DIFFERENCE BETWEEN ANTROCHOANAL AND ETHMOIDAL POLYP



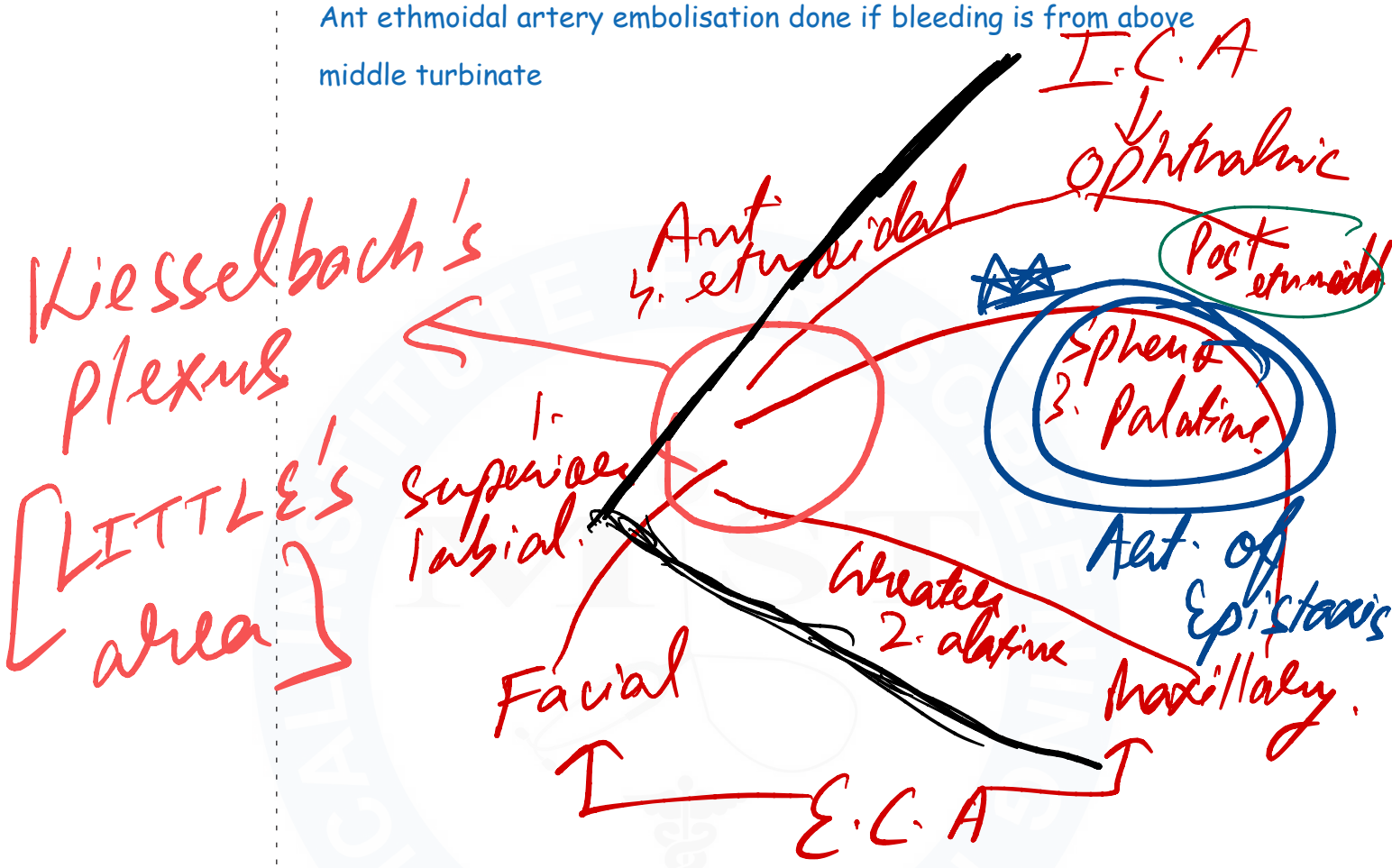
pale, glistening mass that is not sensitive to touch & do not bleed on touch.

Antrochoanal	Ethmoidal
→ grow Posteriorly	→ grows anteriorly.
→ Infective	→ Allergic
→ Children	→ Adults.
→ TT Antibiotics	→ TT Antiallergic MIST

Active space

BLOOD SUPPLY OF NASAL SEPTUM & EPISTAXIS

Ant ethmoidal artery embolisation done if bleeding is from above middle turbinate



M/C/C of Epistaxis in children: Nose picking (Trauma)

M/C/C of Epistaxis in young males: J.N.A
Juvenile Nasopharyngeal

MIST

M/c/c of Epistaxis in Adults: Hypertension > idiopathic

Management of Epistaxis

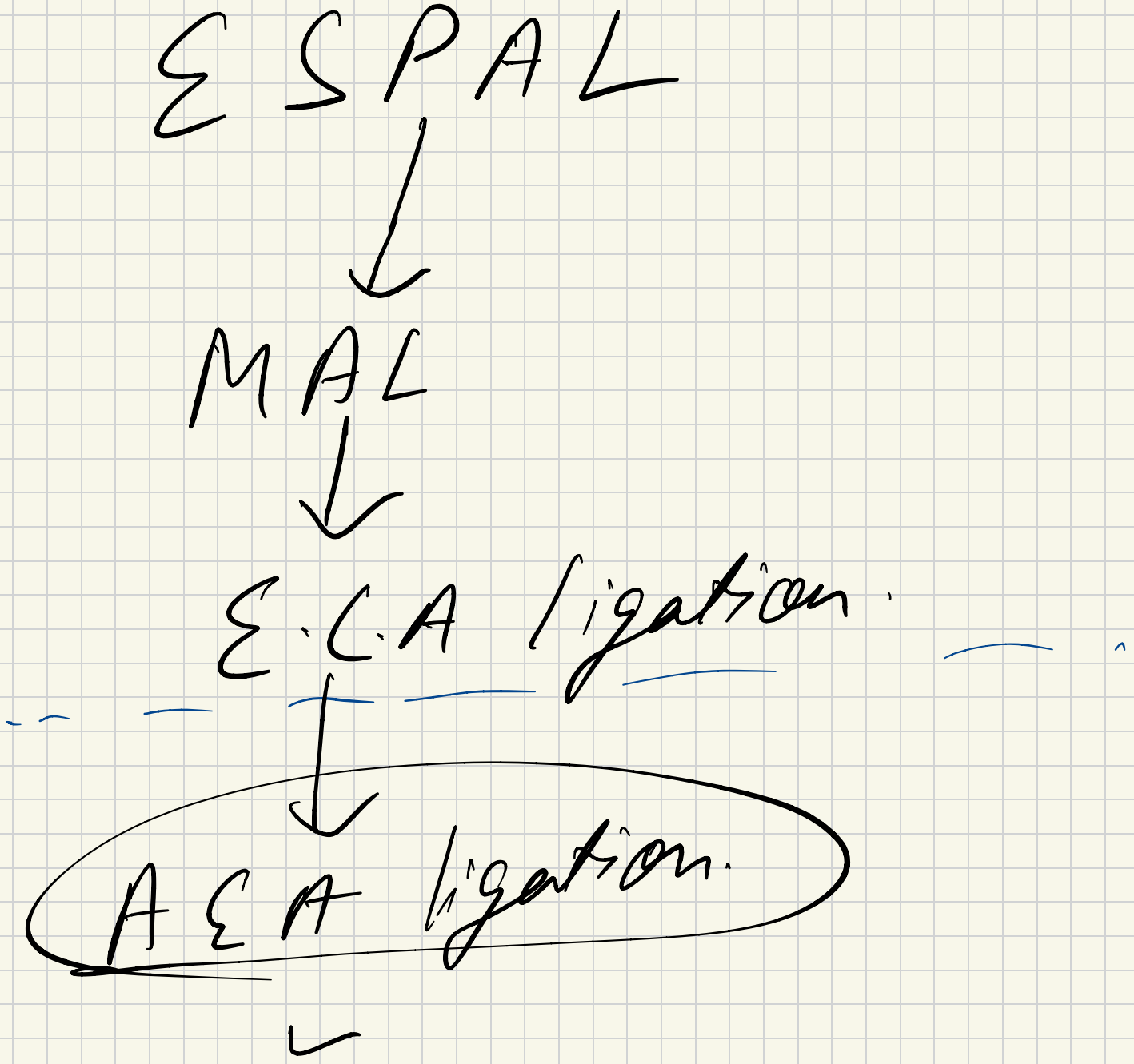
Trotter's Method →

Cold ice packs
+
Xylometazoline.

↓

Nasal packing

↙


ESPAL
MAL
E.C.A ligation.
AEA ligation.

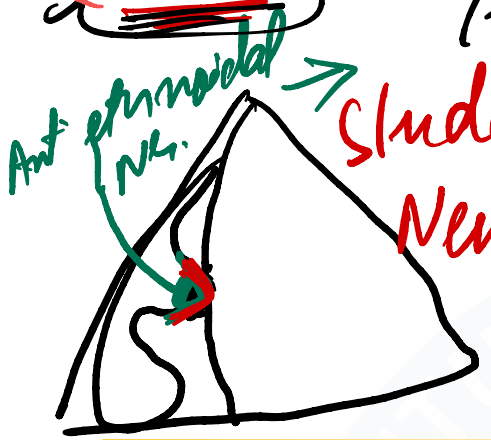
Killian's

Angiofibroma.

DNS

Deviated Nasal Septum.

SEPTOPLASTY V/S SMR (sub mucosal resection)



Sludden's Neuralgia

Birth trauma (M/C)

RHINOPLASTY

Sx of external nose.

1. Depression Of Dorsum- Saddle Nose Cause - Surgical Trauma

   Treatment- Augmentation Rhinoplasty

   - From Iliac Crest

2. Swelling of dorsum - hump nose

   Mainly is congenital

   Dorsal preservation

   Or reduction rhinoplasty

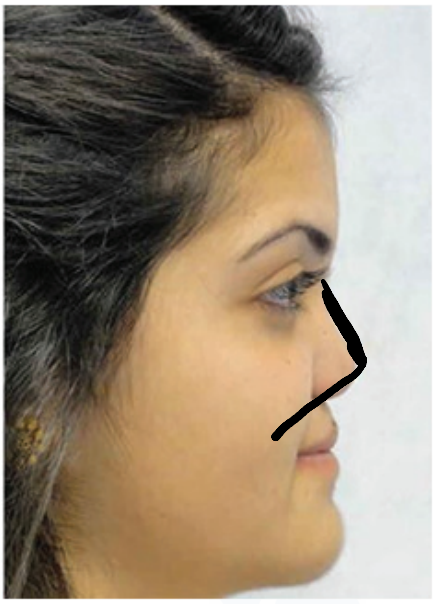
M/c complication of endoscopic sinus surgery is

Synechiae / Adhesion

Treatment - mitomycin c

Antifibrotic drug.

MIST



ENDOSCOPE

Angles of endoscope

Diameter of endoscope- 4mm , 2.7 mm

Adult → Paediatric.

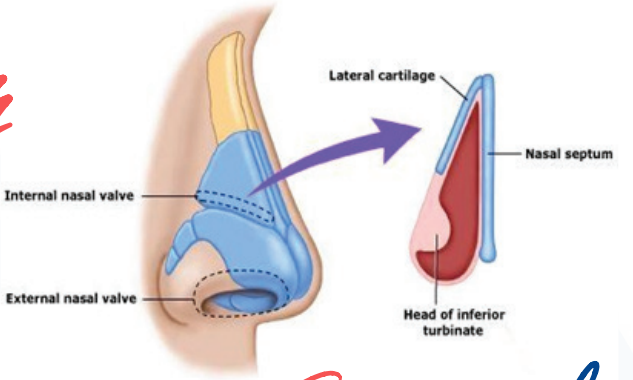
DIAGNOSTIC ENDOSCOPY

A—->P stamberger approach

P—-> A wigand approach

Active space

INTERNAL NASAL VALVE / LIMEN NASI



Patency

Cottle test [Nasal valve patency test]

Septal Haematoma →

Rhinolalia clausa - hyponasality → decreased nasal component → nasal obstruction

Rhinolalia aperta → hypernasality →increased nasal component → palate insufficiency

Active space

X-ray PNS

Waters view / o.M view → best for maxillary sinus

= Occipitomental view

Pieres view(w+open mouth) — for sphenoid sinus

Caldwell view — for frontal and ethmoid sinus

Testosterone dependent

1. Exclusively seen Boys
2. Profuse Epistaxis

JNA ⇒ Juvenile Nasopharyngeal Angiofibroma.

3 ⇒ Sphenopalatine foramen SPA

Hollman miller sign ⇒ Pushing the maxillary sinus walls anteriorly.

Dumbell shaped tumour-pterygopalatine fossa→infratemporal fossa

PPF — S.PF — I.T.F

MIST

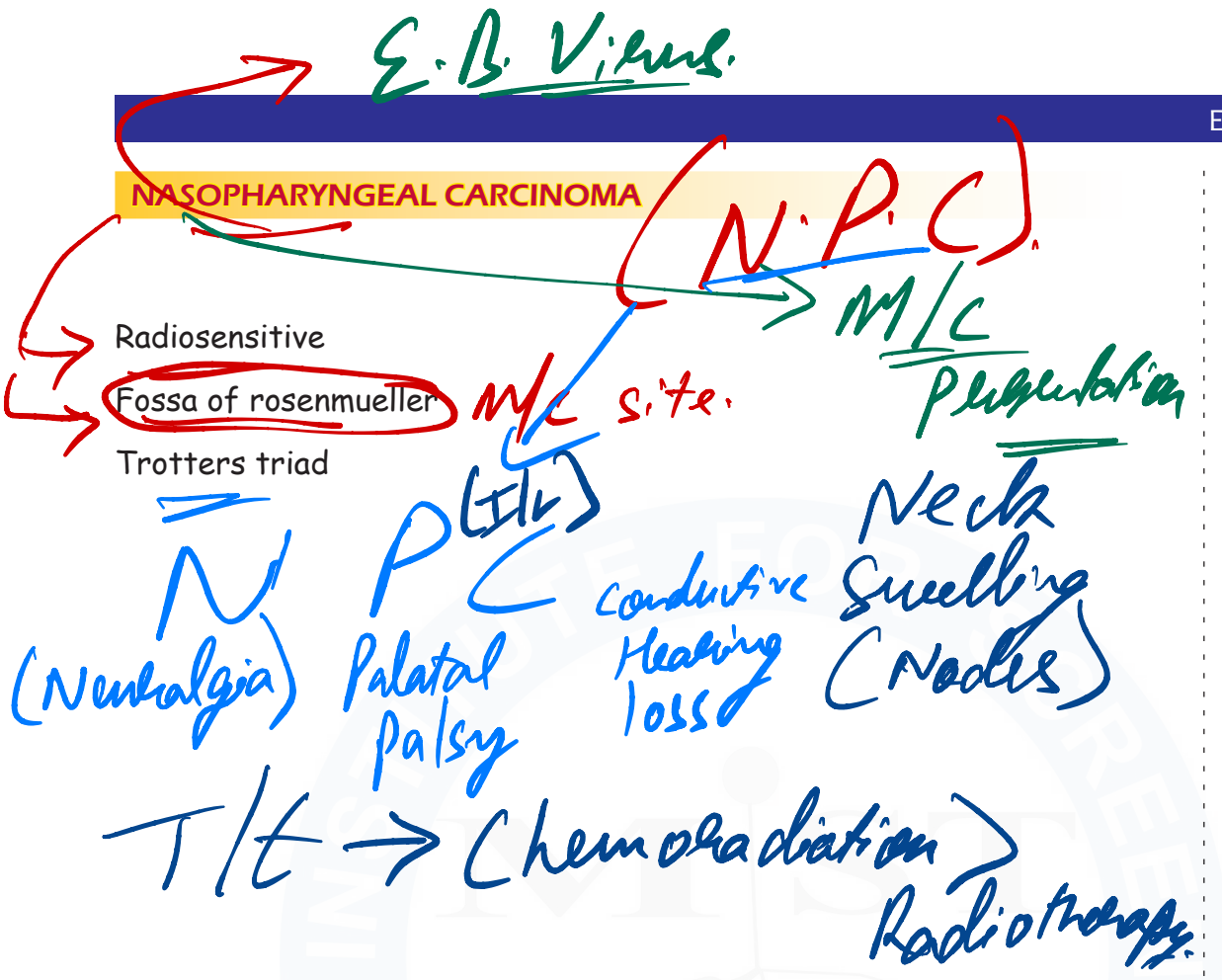
Activate space

→ Frog face deformity.
(facial swelling).

→ Radowska staging.

→ Ex. (T/E)

NASOPHARYNGEAL CARCINOMA



Active space

Workbook Question

1.* A housewife presents with complaints of a blocked nose, loss of smell, and dry, greenish crusty nasal discharge. On examination, nasal cavities appear roomy and turbinates are atrophied. Which of the following surgeries can be done on this patient?

A. Submucous resection
B. Young's operation
C. Septoplasty
D. Vidian neurectomy

2. A 23 year old male presents to you with a nasal discharge which has the following appearance on a handkerchief. Presence of which of the following features helps you in diagnosing the condition as CSF rhinorrhoea?

A. Halo sign
B. Handkerchief sign
C. Schwartz sign
D. Phelps sign

3.* What clinical test is depicted in the image?

A. Epley's maneuver
B. Heimlich maneuver
C. Cottle's test
D. Trotter's method

Active space

4. A 14 year old boy who is a football player comes to the casualty with complaints of epistaxis while playing football on the ground. There is no history of trauma or fall. He also gives a history of recurrent spontaneous nasal bleeding and also unilateral nasal obstruction and nasal speech. Contrast-enhanced CT scan of the head is shown below what is this sign known as?

A. Aquino Sign
B. Holman Miller sign
C. Teardrop sign
D. Hanging drop sign

5. Office headache is due to inflammation of which sinus?

A. Frontal
B. Maxillary
C. Ethmoidal
D. Sphenoid

6. * Gold standard investigation before FESS (functional endoscopic sinus surgery) is?

A. MRI scan
B. Xray
C. Antral lavage
D. CT scan

7. *Which of the following is part of trotter's triad, seen in nasopharyngeal carcinoma?

A. Conductive hearing loss, ipsilateral temporoparietal neuralgia, Palatal paralysis
B. Sensorineural hearing loss, contralateral temporoparietal neuralgia, palatal paralysis
C. Sensorineural hearing loss, ipsilateral temporoparietal neuralgia, palatal paralysis
D. Conductive hearing loss, contralateral temporoparietal neuralgia, palatal paralysis

Active space

8. * Which of the following statements is true with respect to the Little's area?

A. Keisselbach's plexus is present in the Little' area
B. Itis situated at the postero-inferior part of the nasal septum
C. Least common site for epistaxis
D. Anastomoses of 7 major arteries occur here

9. * What is this view called in which maxillary sinus is seen best?

A. Water's view
B. Towne's view
C. Law's view
S. Stenver's view

10. * An elderly diabetic female who was treated with steroids and remdevisir for COVID now presented with a black, foul-smelling discharge from the nose. On examination, blackish necrotic mass filling the nasal cavity and eroding the septum and hard palate. What is the most probable diagnosis?

A. Aspergillosis
B. Mucormycosis
C. Foreign body
D. Rhinosporidiosis

11. *Identify the sinus marked in the image.

A. Frontal sinus
B. Pyriform sinus
C. Ethmoid sinus
D. Maxillary sinus

Active space

12. *What is this view called in which frontal sinus is seen best?

A. Caldwell's view
B. T owne's view
C. Law's view
D. Stenver's view

13. *A 60-year-old male presents with painless cervical lymphadenopathy. On examination, the right ear reveals conductive hearing loss with a dull tympanic membrane. Moreover, decreased mobility of the soft palate was also noted. What is the probable diagnosis?

A. Nasopharyngeal carcinoma
B. Adenoid cystic cancer
C. Juvenile nasopharyngeal angiofibroma
D. Quinsy

14. *Which of the following fungal infection is most commonly seen in the maxillary sinus?

A. Blastomycosis
B. Aspergillosis
C. Candidiasis
D. Histoplasmosis

15. Which of the following is true regarding the external nose?

A. Bone 1/3rd Cartilage 2/3rd
B. Bone 2/3rd Cartilage 1/3rd
C. Bone 1/2rd Cartilage 1/2rd
D. Bone 1/3rd Cartilage 1/3rd

Active space

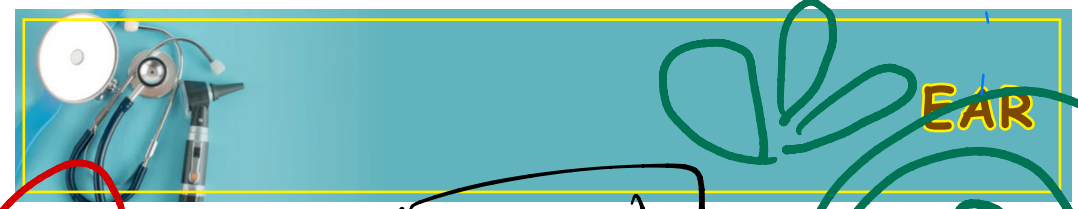
16. A 35-year-old female takes a tablet of aspirin for her headache, later she presented with wheezing and breathlessness. These symptoms are most likely due to which of the following?

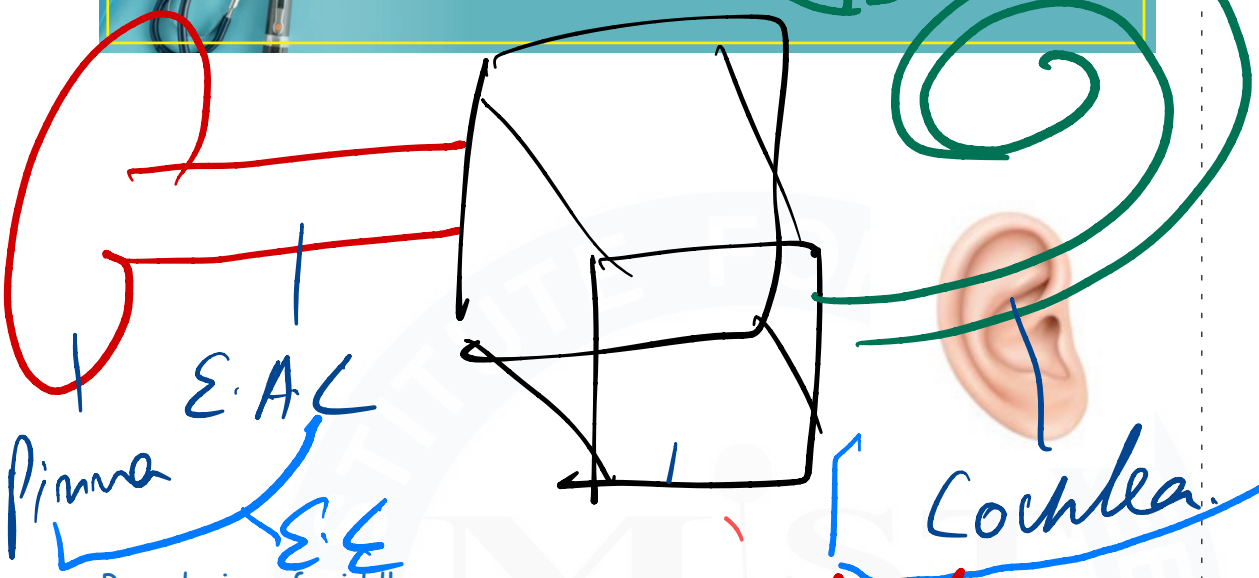
A. I gE release
B. Nasal polyp
C. Drug interaction
D. Extrinsic asthma

17. The patient presents with complains of unilateral nasal obstruction associated with watery nasal discharge or nasal bleeding. And the diagnosis of Inverted Papilloma is made. Which is true regarding the following condition?

A. M alignant
B. Benign and non-invasive
C. Benign and invasive
D. None of the above

Active space





Boundaries of middle ear

Lat. → Tympanic membrane.
Floor → Jugular bulb.
Roof → Tegmen tympani
Ant. → I.C.A, E.tube, Tensor tympani
Med. → Promontory, O.W, R.W, Horizontal F. Nerve.
Post → Mastoid, Vertical F. Nerve, chorda tympani, Facial recess, Sinus tympani

EAR FORMATION

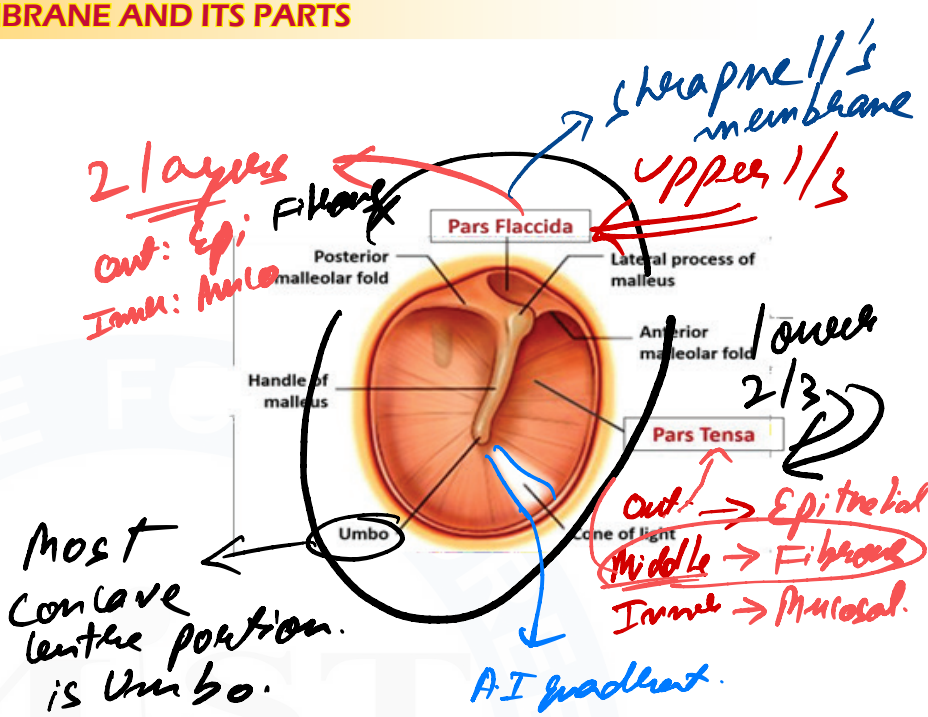
1st branchial arch (ectoderm)--> ext. Ear

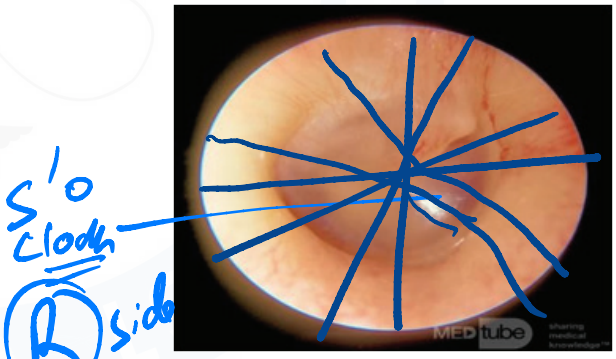
2nd branchial arch(endoderm)--> middle ear

Otic capsule(neuroectoderm)--> inner ear

Activ space

TYMPANIC MEMBRANE AND ITS PARTS





10 mm DIAMETER

0.1 mm

thickness 15 mg weight

CONE OF LIGHT CLOCK POSITION

T M MARKERS →

Concave in shape Umbo(best marker)

Cone of light -ai quadrant

Handle of malleus

REFERRED OTALGIA

Oral cavity, dentition → Trigeminal Nv.

Oropharynx, Tonsils → Glossopharyngeal

Larynx → Vagus.

NERVE SUPPLY OF TM

V3 (mandibular div. of Trigeminal Nr.)

Outer surface

Ant half - auriculotemporal nerve

Post half - auricular branch of vagus

↓↓

ARNOLD's Nr.

Inner surface

Tympanic branch of glossopharyngeal nr.

↓↓

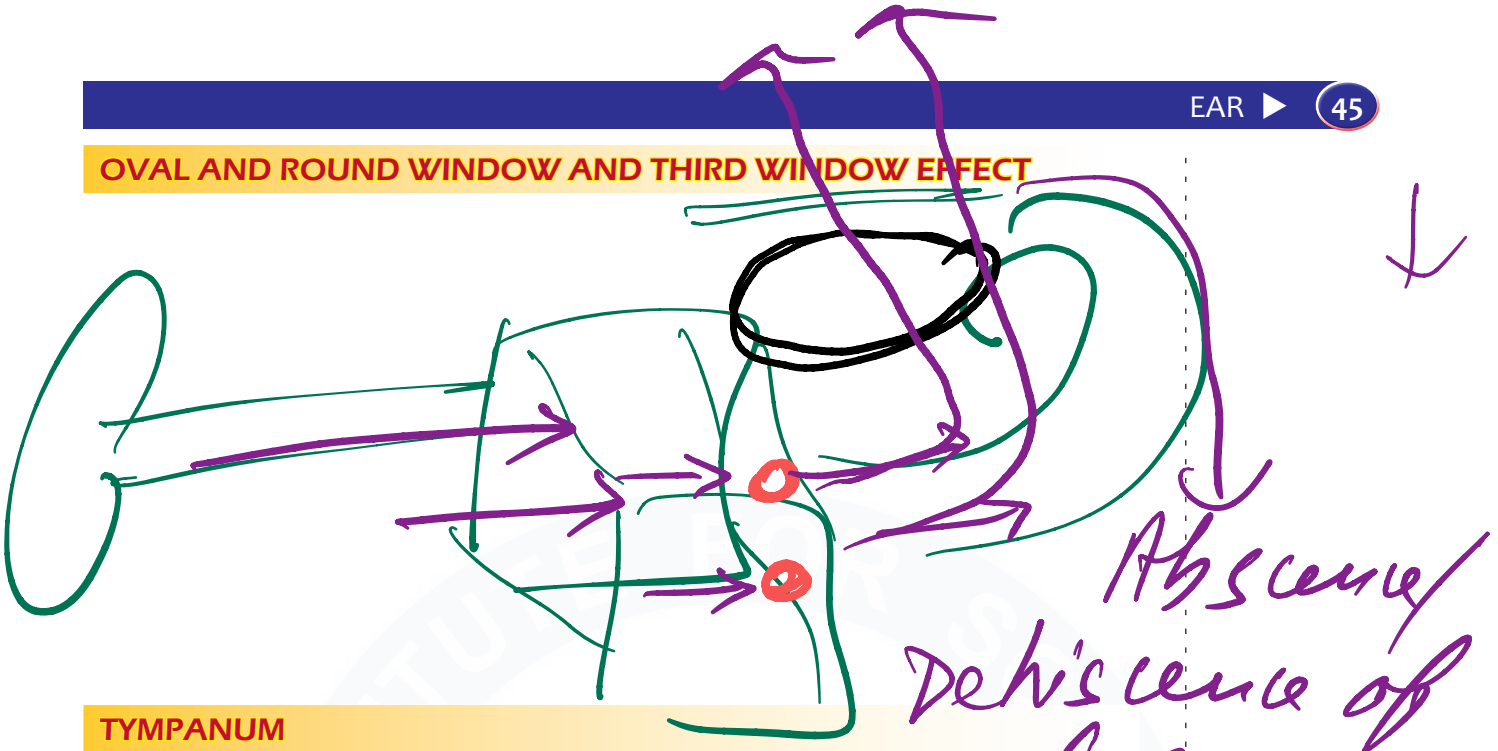
JACOBSON's Nr.

ARTERIAL SUPPLY

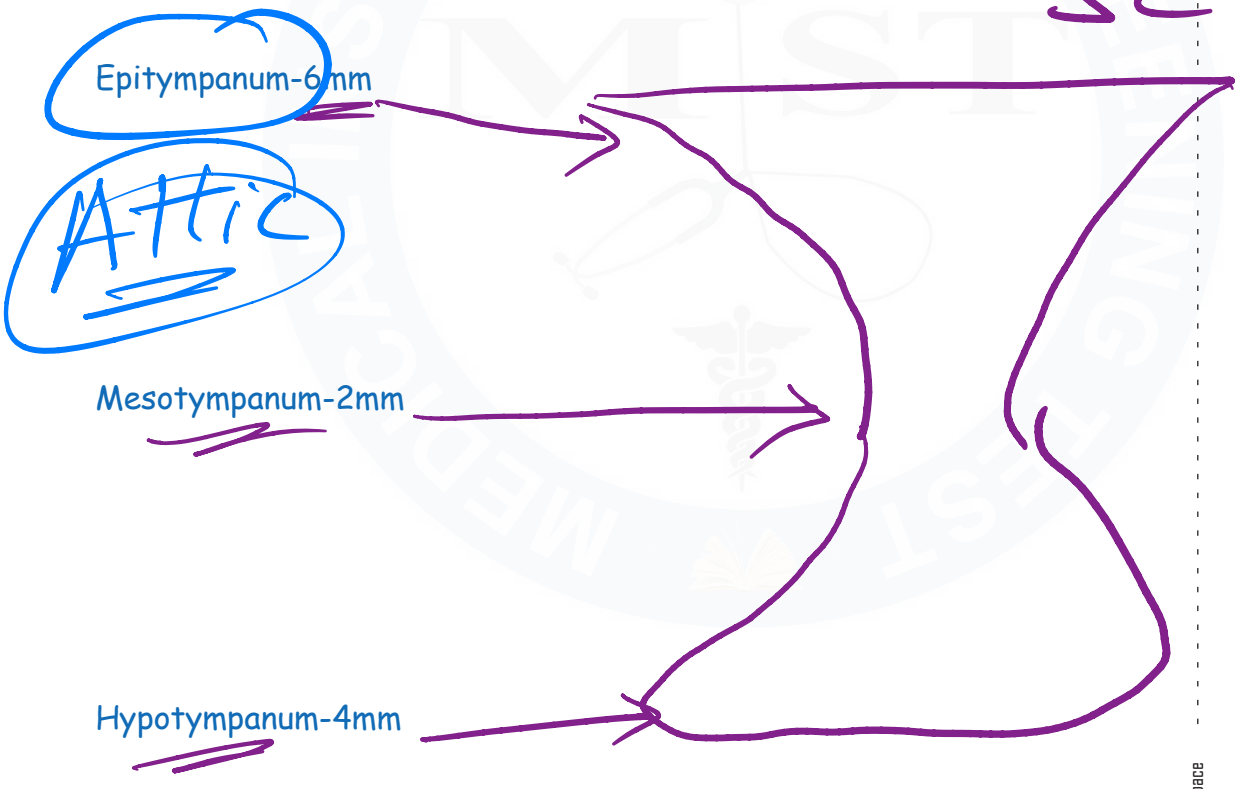
Maxillary artery

Active space

OVAL AND ROUND WINDOW AND THIRD WINDOW EFFECT

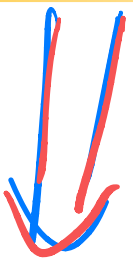


TYMPANUM

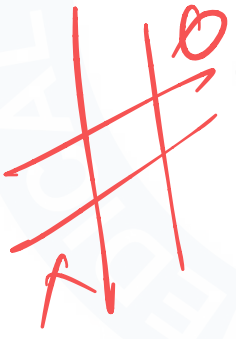


Active space

PROTYMPANUM

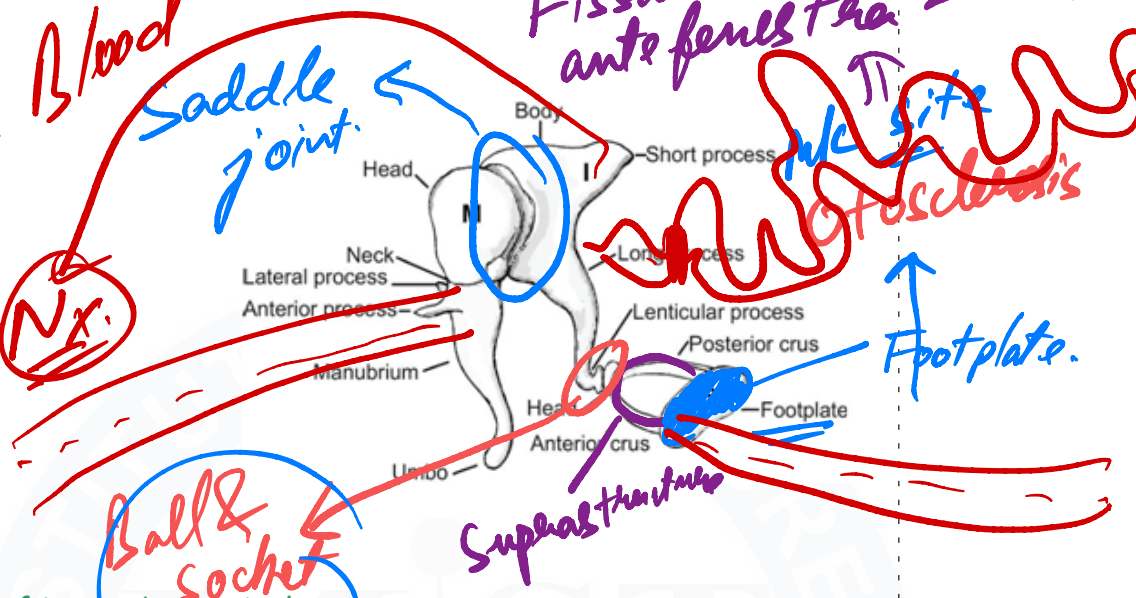


Part of middle ear surrounding eustachian tube



Active space

OSSICLES AND ITS PARTS



Long process of incus aka lenticular process

Lenticular process is m/c site of necrosis

Is joint is ball and socket joint → most prone to trauma

Foot plate of stapes → m/c site of otosclerosis

Active space

DEVELOPMENT OF EAR OSSICLES

Malleus and incus develop from 1st branchial arch

Stapes from 2nd branchial arch

Except medial surface/foot plate that develops from otic capsule

Oval window-aka - fenestra vestibuli

Round window - aka - fenestra cochlea

Active space

NERVE SUPPLY OF MIDDLE EAR

Sensory

9th nr. Jacobson

10th nerve- arnold

Together form tympanic plexus

Motor supply

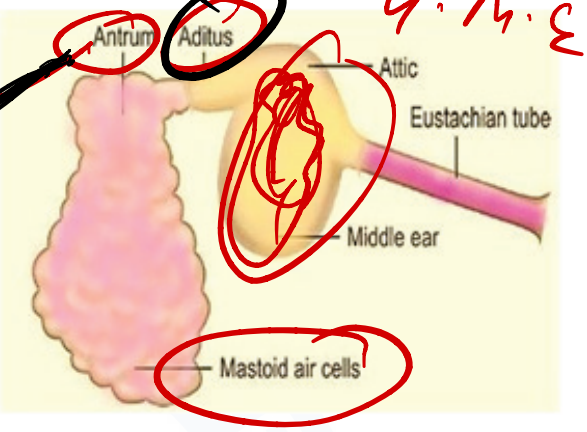
Tensor tympani supplied by mandibular division of trigeminal nr.

Stapedius by 7th nr.

Dampen loud sound. (Protect against loud Noise/ sound)

Active space

WHAT IS MIDDLE EAR CLEFT



Thickness of mastoid in adult is 15 mm

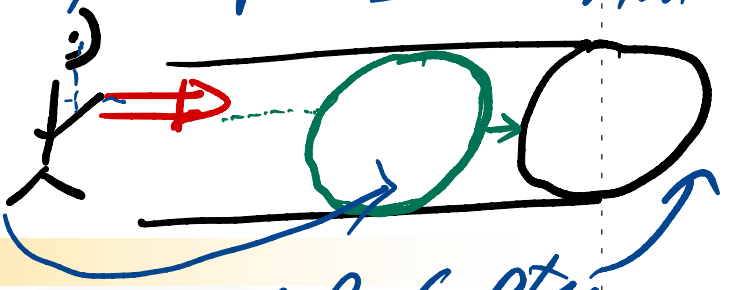
Thickness of mastoid in newborn is 1-2 mm

Grows by the rate of 1-2 mm per year so in 15 years reaches adult thickness

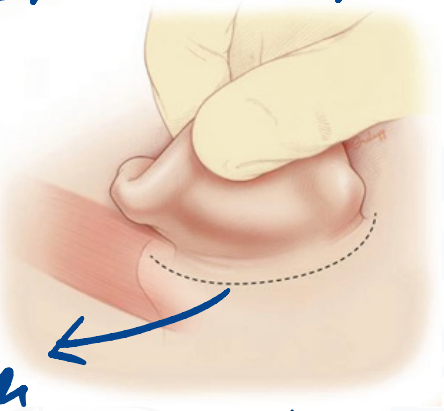
Tip is absent in newborn where facial nr. Is running so it is exposed

KORNER SEPTUM

Petrosquamous suture line



TO REACH MIDDLE EAR



Mastoid

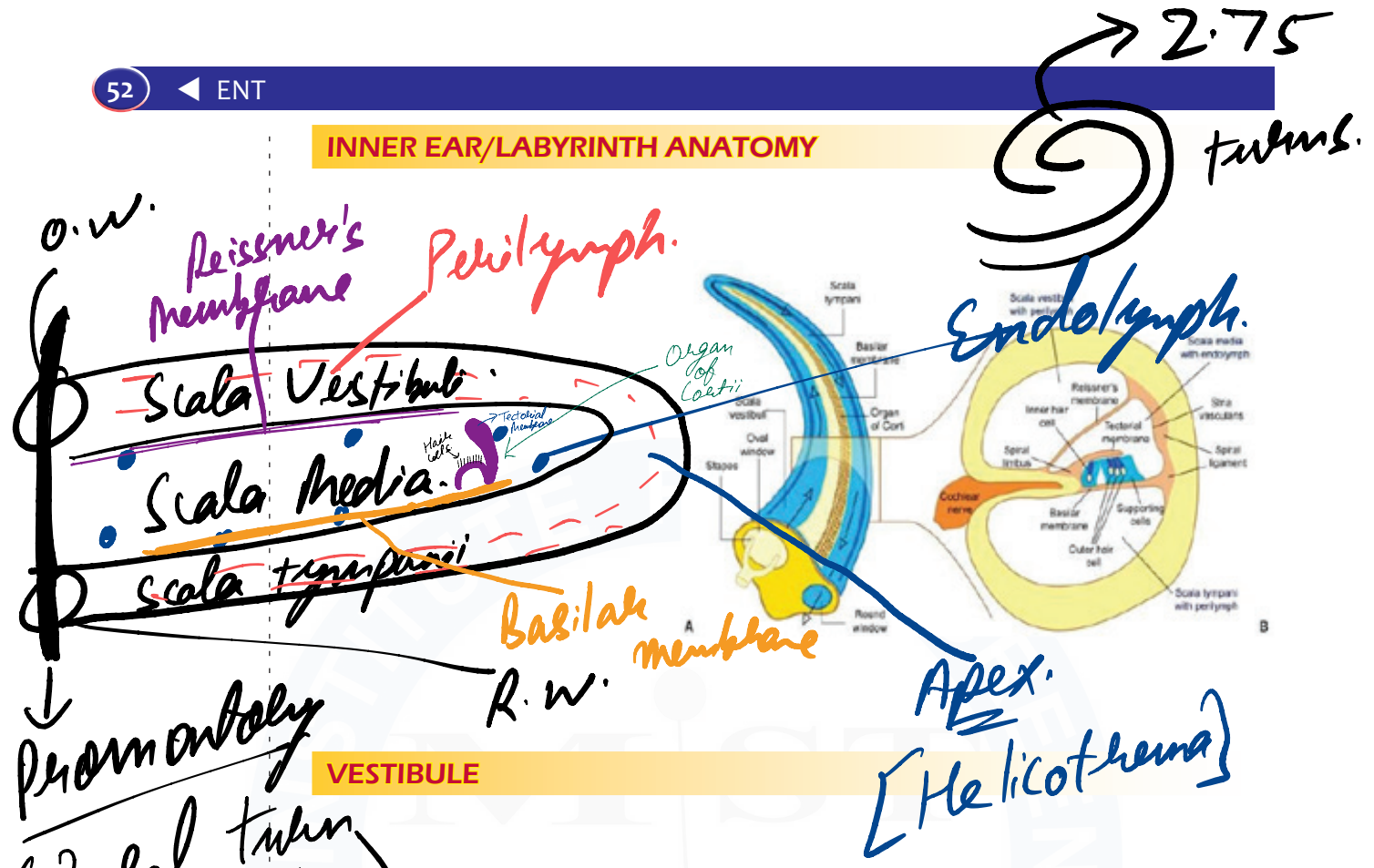
(post auricular approach)-wildes incision

Annulus

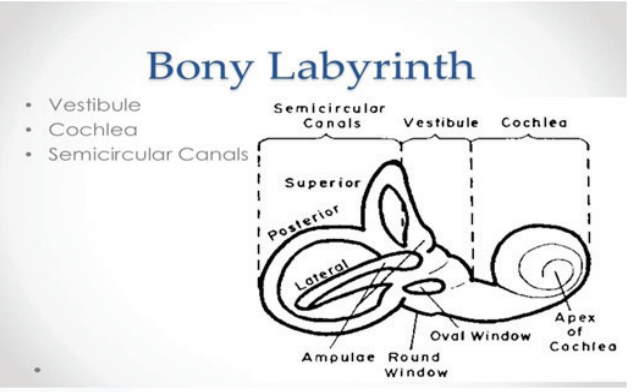
(Endomeatal approach)-rosen incison

MIST

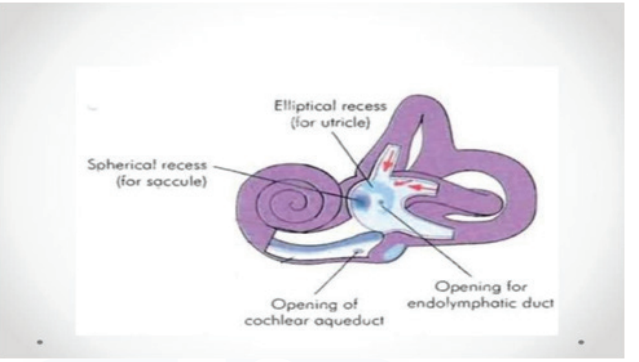
INNER EAR/LABYRINTH ANATOMY

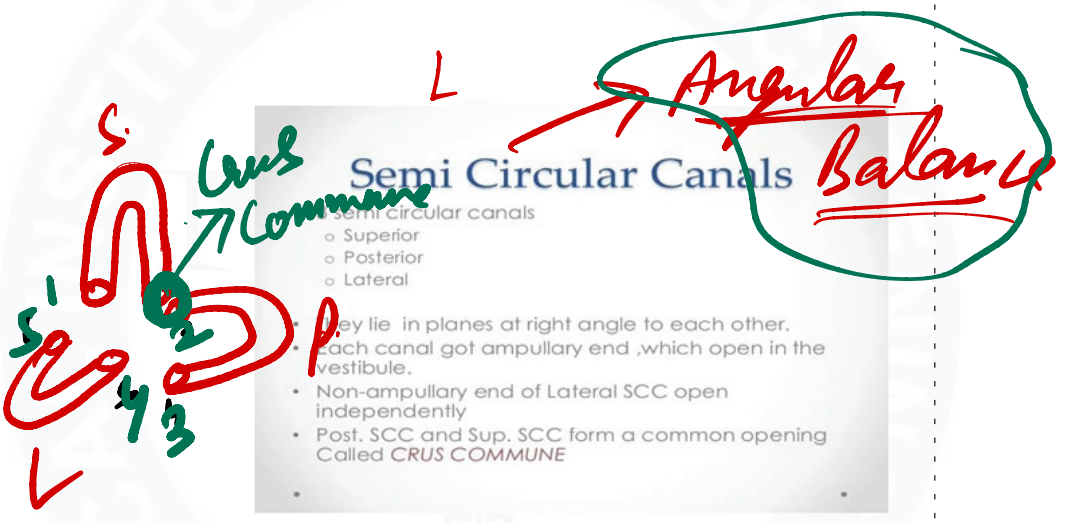


VESTIBULE

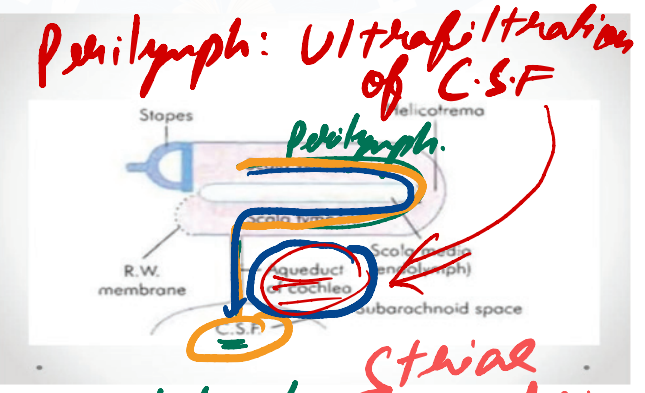


Active space



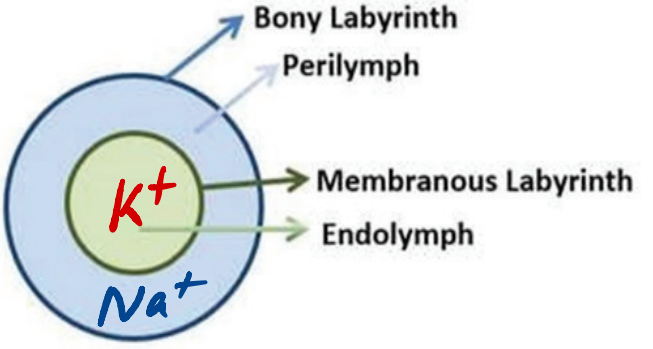


1. Perilymph: Ultrafiltration of C.S.F



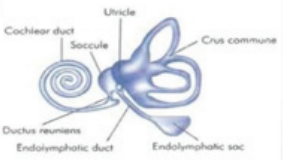
2. Endolymph: Striae Vascularis

Active space



Membranous Labyrinth

- It lies within the bony labyrinth
- Filled with Endolymph.
- Space between membranous labyrinth and bony labyrinth- Perilymph.
- It contains
  - Cochlear Duct
  - Utricle and Saccule(otolith organs)
  - Semi circular Duct
  - Endolymphatic Duct and Sac



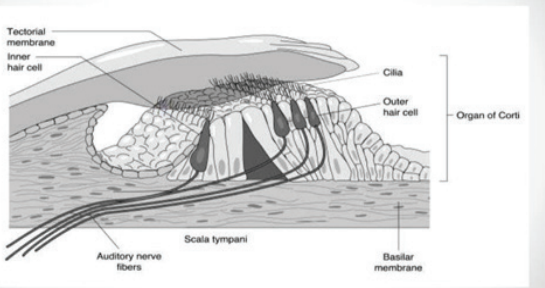
ORGAN OF CORTI

- Organ of hearing.
- Situated on basilar membrane of cochlea

- 3 parts
  - Sensory Hair cells
  - Supporting cells
  - Tectorial Membrane

Sound energy converts into Electrical impulse

Active space



Inner Hair cells	Outer Hair cells
• Total No : 3500	• 12,000 [1:4 ratio]
• Arranged in single row	• Multiple rows
• Flask shaped	• Cylindrical
• 95% afferent from Cochlear nerve (8th CN)	• Efferent from Sup. Olivary complex
• Stereocilia arranged in V or W format	• Stereo cilia single continous layer.
• More resistent	• Easily damaged to drugs & loud sound
• Transmit auditory stimuli	• Modulate the IHC.

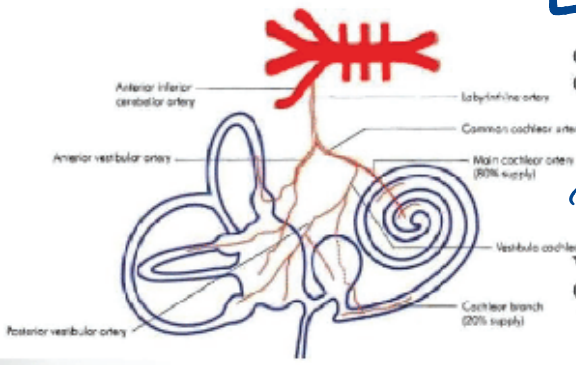
Supporting cells

- Stabilize and support the organ of corti
- Supply nutrient to the highly specialized sensory cells(inner and outer hair cells)
- Includes
  - Hensen's Cell
  - Claudius Cells
  - Deiters Cells
  - Boettcher's Cell

Active space

Blood Supply

Arterial



Ant. Inferior Cerebellar Art

Labyrinthine/Internal Auditory art.

Common Cochlear

Ant. Vestibular A (to utricle & LSCC, SSCC)

Vestibulo Cochlear

Main Cochlear 80%

Cochlear branch (20%)

Post. Vestibular A (to saccule & PSCC)

Active space

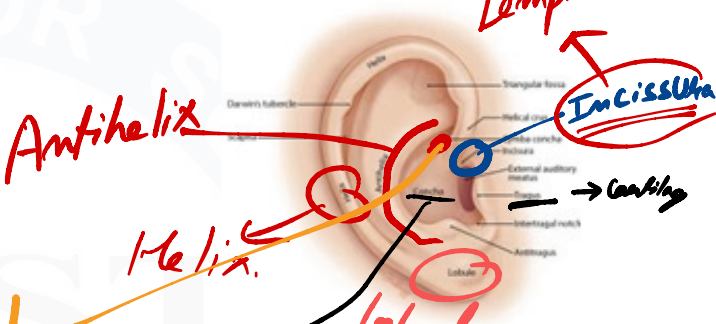
MIST

PART	FUNCTION	DAMAGE	SENSORY ORGAN	
COCHLEA	HEARING	DEAFNESS	ORGAN OF CORTI	
UTRICLE/	LINEAR	ATAXIA	MACULAE	
SACCULE	BALANCE			
SUP. SCC	ANGULAR	NYSTAGMUS	CRISTAE	
POST. SCC	BALANCE	VERTIGO	CUPULAE	
HORI. SCC				

Active space

CONGENITAL ABNORMALITIES OF COCHLEA

EXTERNAL EAR ANATOMY



HILLOCKS OF HIS COMBINE TO FORM PINNA

Incomplete fusion → Preauricular sinus

Active space

OTHER ABNORMALITIES OF PINNA

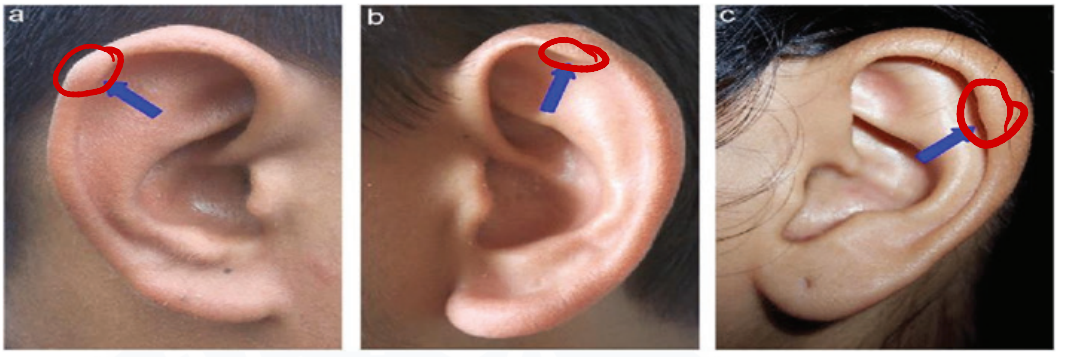
Bat ear → absent anti helix

Wildermuth ear → absent helix

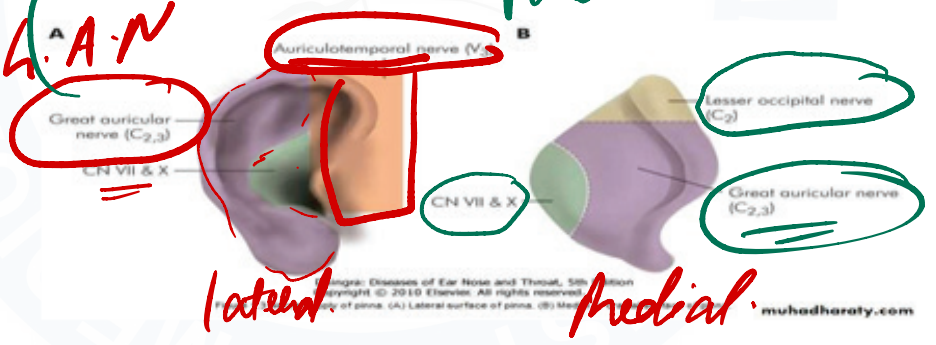
Mozart ear -mixing of helix and antihelix

Active space

DARWIN TUBERCLE



NERVE SUPPLY OF PINNA



Pinna achieve adult size by 6 years of age so never operate before that

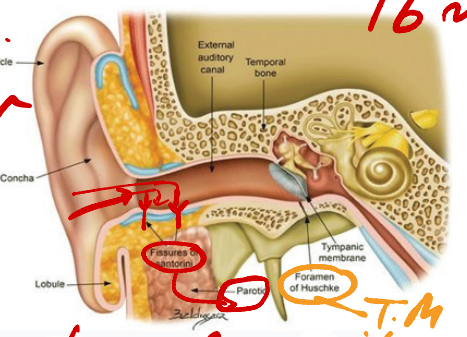
Active space

EAC

Length Isthmus

Organic>>> inorganic

Most dangerous - button battery —> leads to necrosis of surrounding tissue and facial palsy



Fissures of santorini- normal (parotid)

Foramen of huschke- abnormal (tm joint)

EUSTACHIAN TUBE

Length

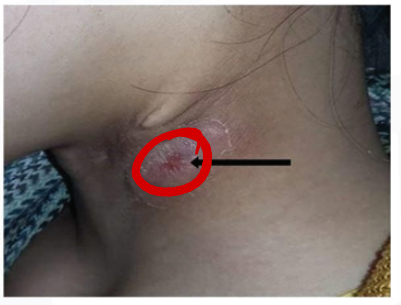
Active space

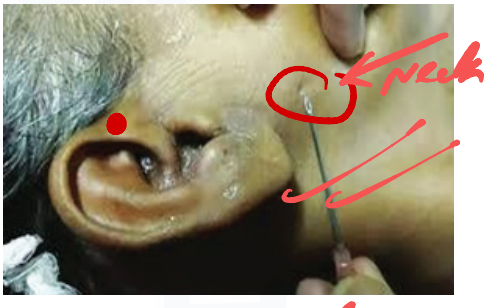
- MIST

COLLAURAL FISTULA

Abnormality of the first branchial cleft One opening in the floor of eac

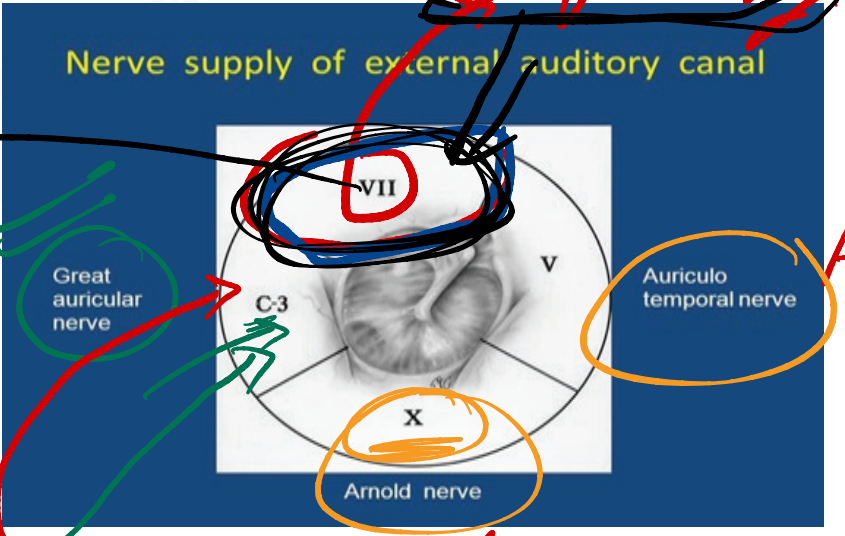
Second opening between scm and mandible in neck/ sometimes behind the mandible





T/t → Fistulectomy.

NERVE SUPPLY OF EXTERNAL AUDITORY CANAL



Active space

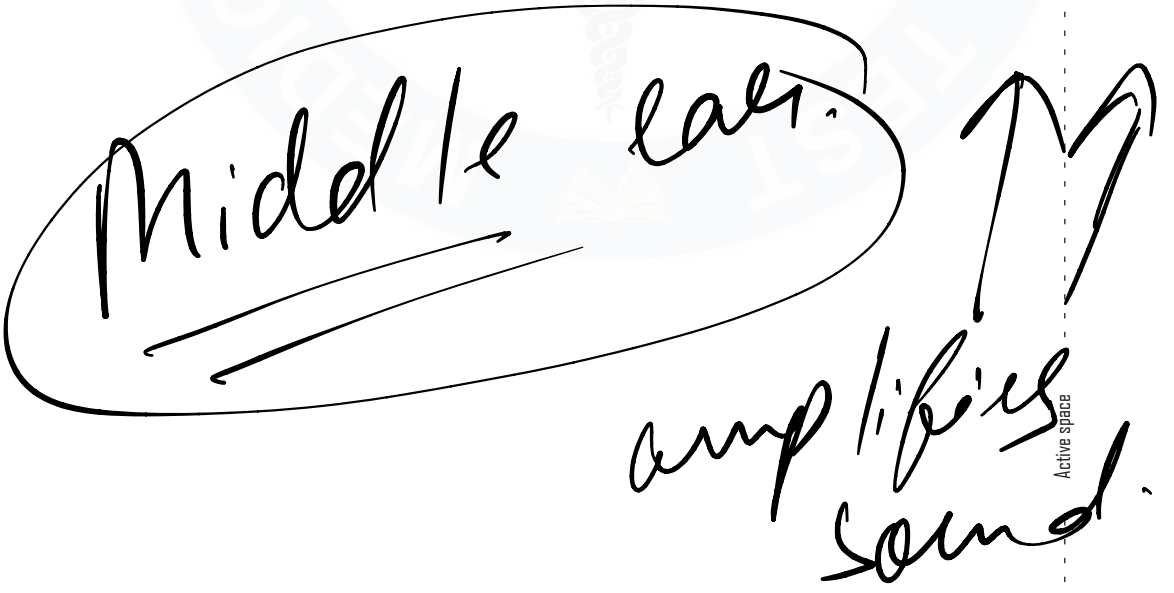
CLINICAL ASPECT

Nr. Of wrisberg(vii nr.) And hitzelbergerger sign

Cough reflex in ai part due to vagus

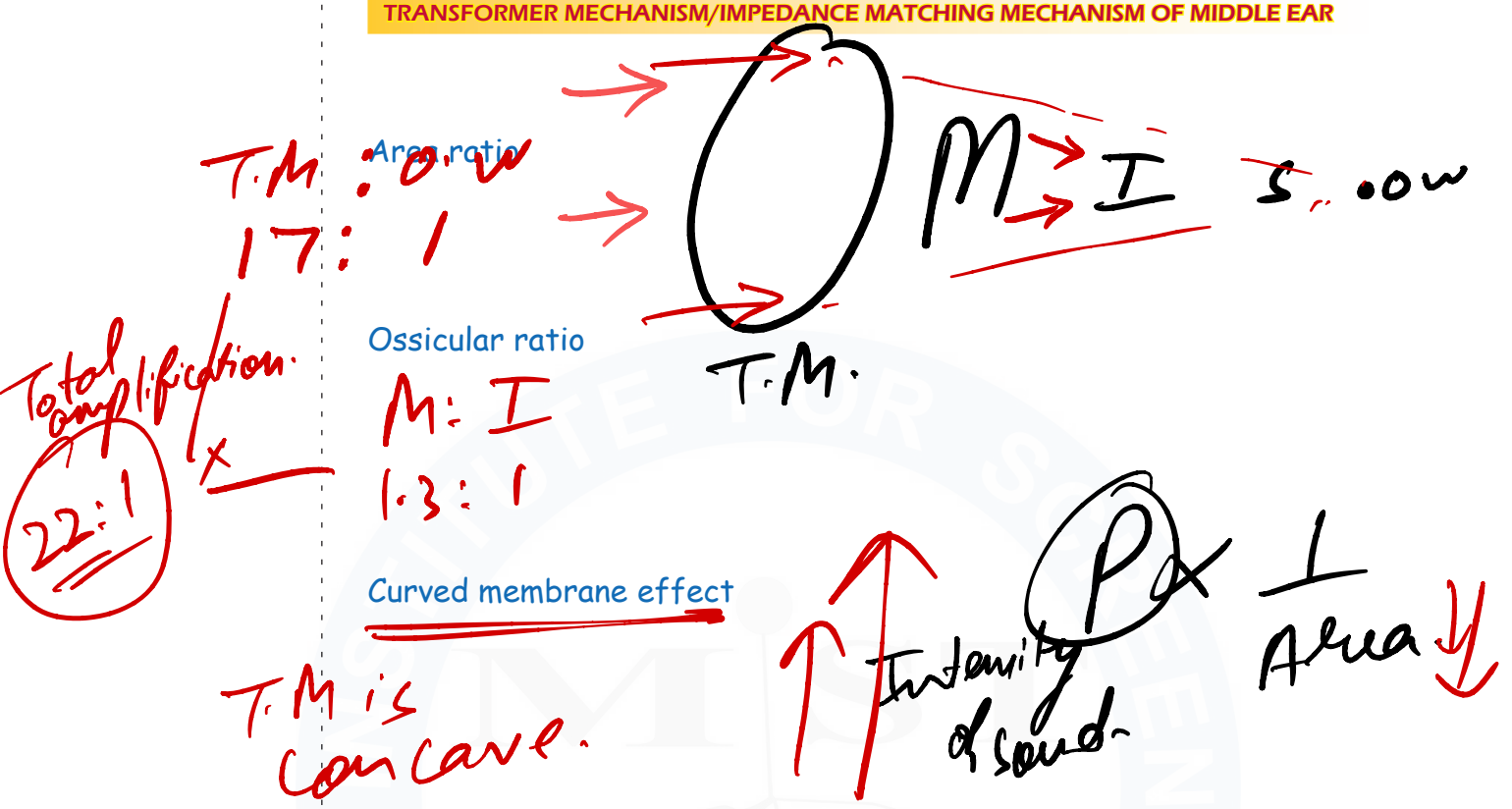
Loss of sensation in pi part due to tb spine

PHYSIOLOGY OF HEARING AND AMPLIFICATION

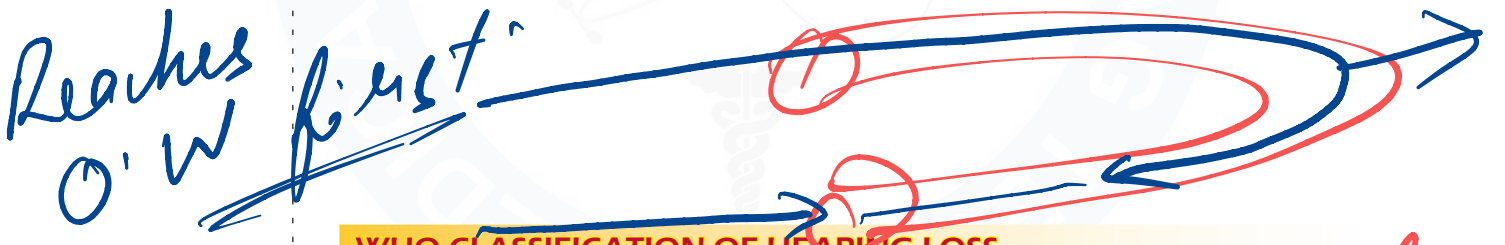


Active space

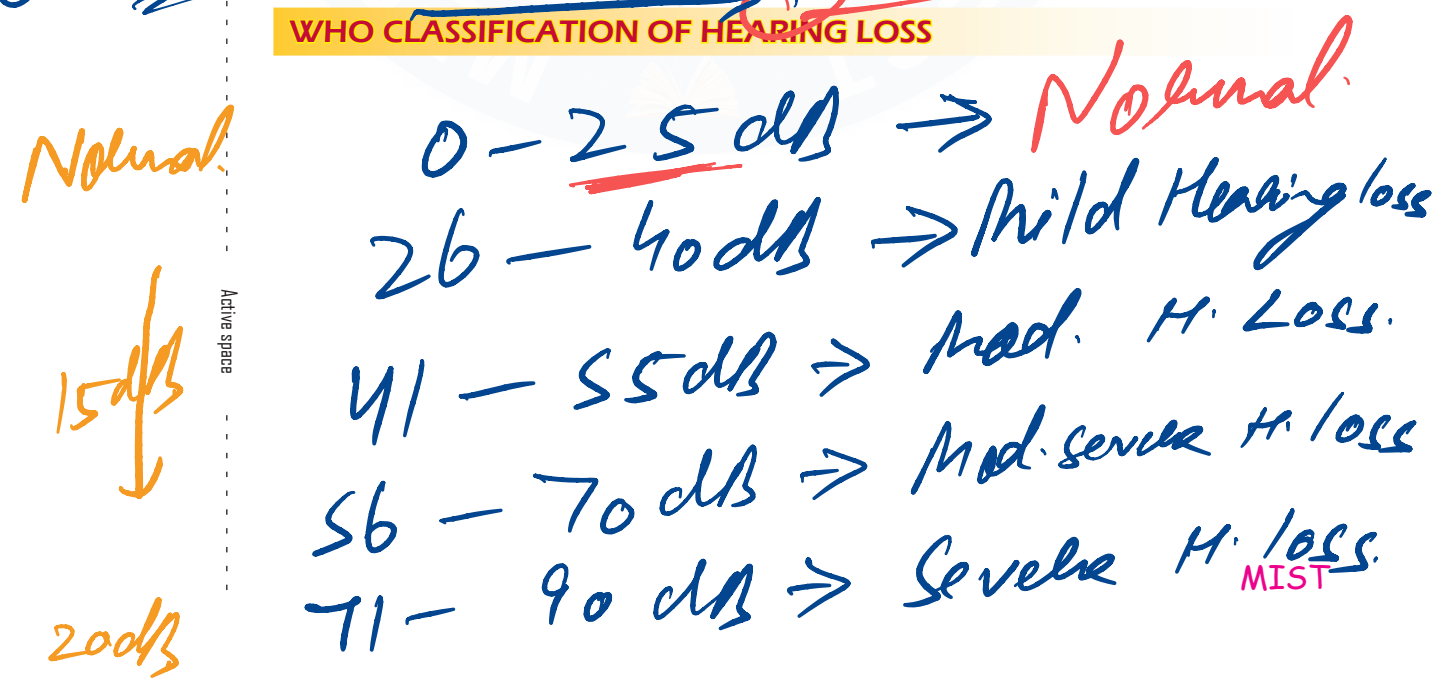
TRANSFORMER MECHANISM/IMPEDANCE MATCHING MECHANISM OF MIDDLE EAR



Phase difference of the two windows



WHO CLASSIFICATION OF HEARING LOSS



Active space

AUDIOMETRY

Frequency(hz)

Intensity(db)

Audiometric zero-25db

DEAFNESS- PT RESPONDS >25db

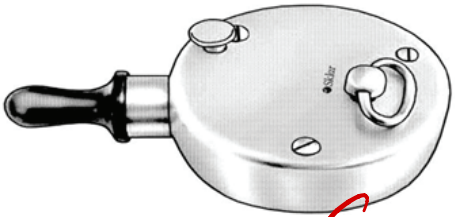
If no response at all- dead ear Masking

Masking done using barany noise box

DEAFNESS- PT RESPONDS >25db

If no response at all- dead ear Masking

Masking done using barany noise box



Pink noise /sound- mainly consistent single frequency

White sound/noise- multiple frequency sound(sort of wide band)

MIST

Active space

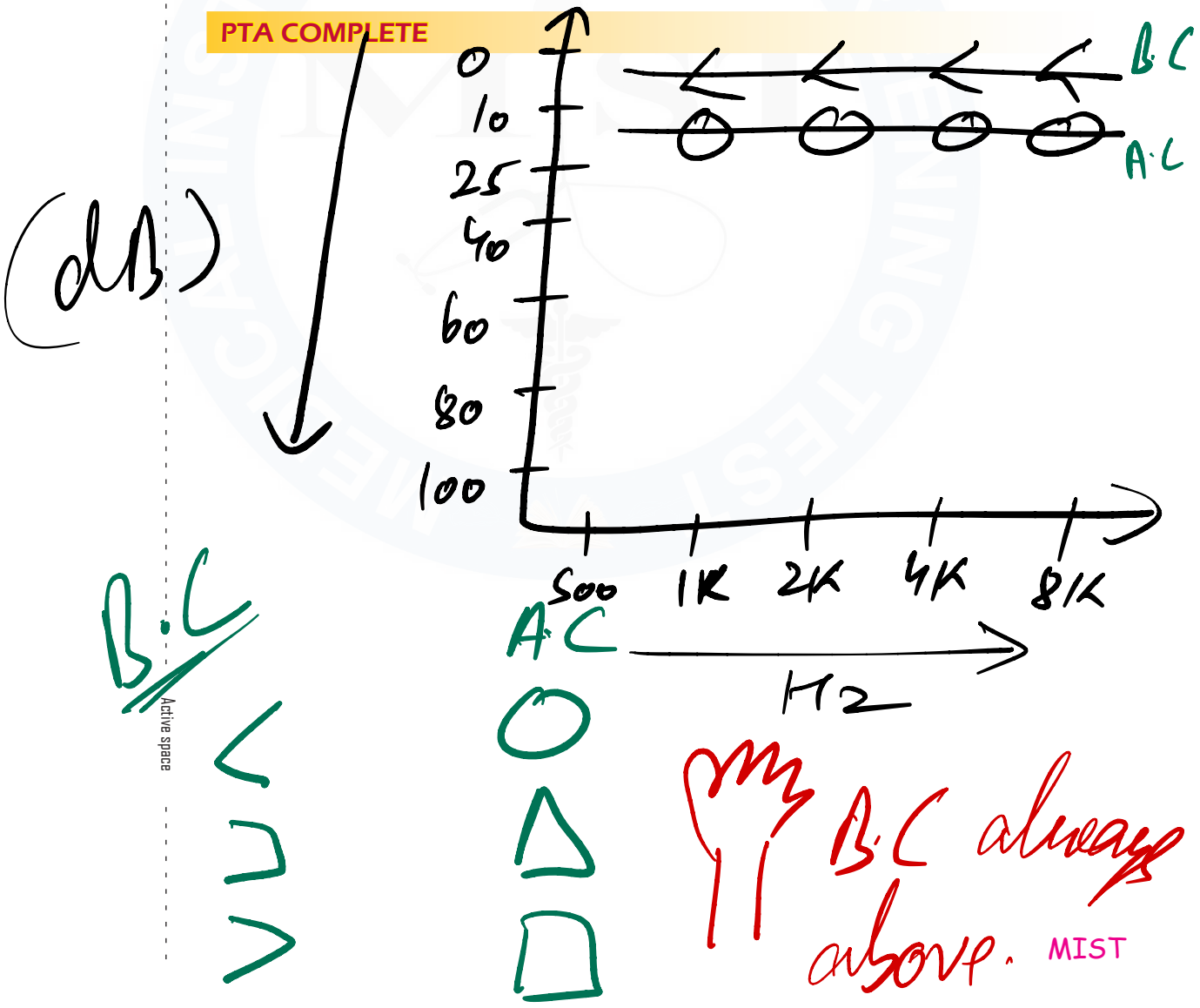
LEVELS TILL WHICH YOU WILL SEE

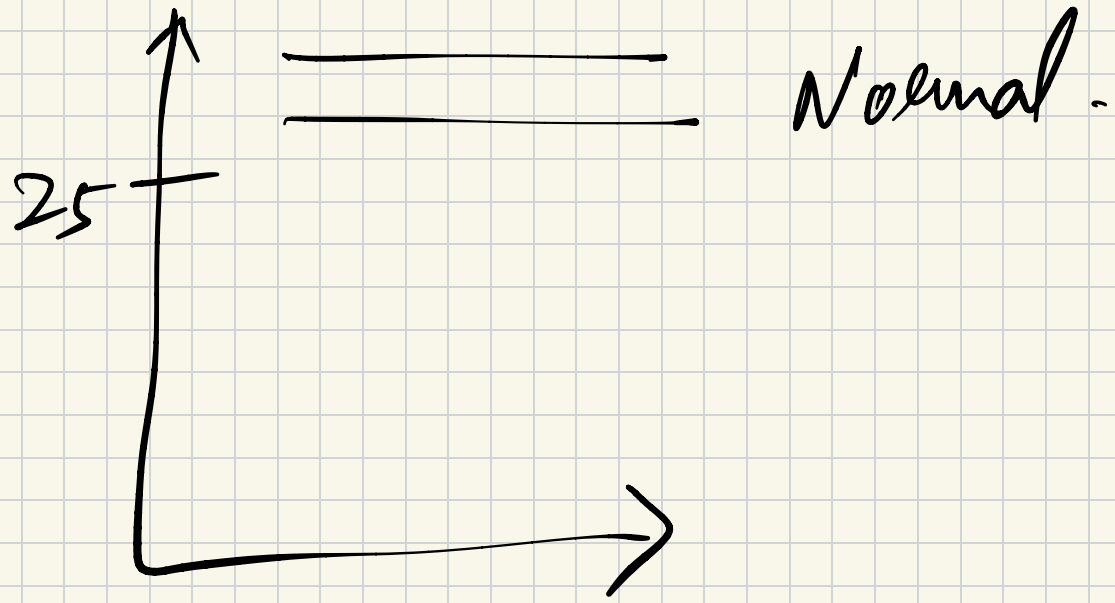
Conductive deafness - Ext. And middle ear

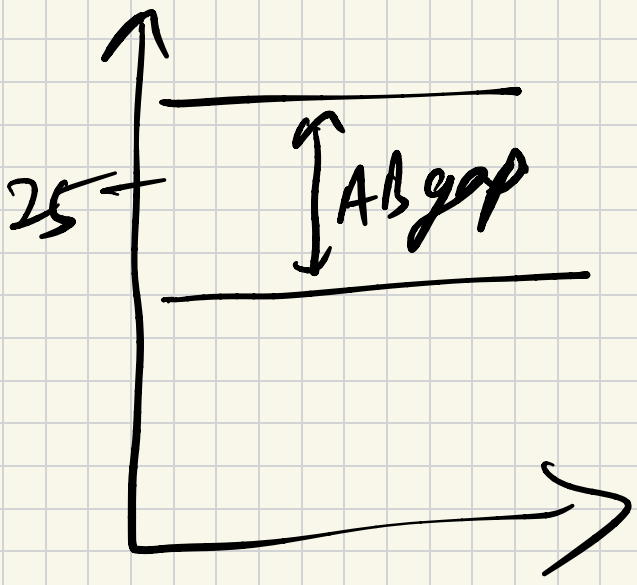
Sensory deafness - Cochlea

Neural deafness - Retrocochlear (8th nr and beyond)

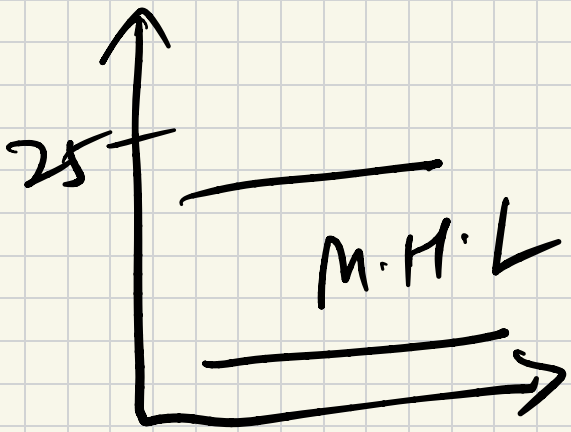
PTA COMPLETE

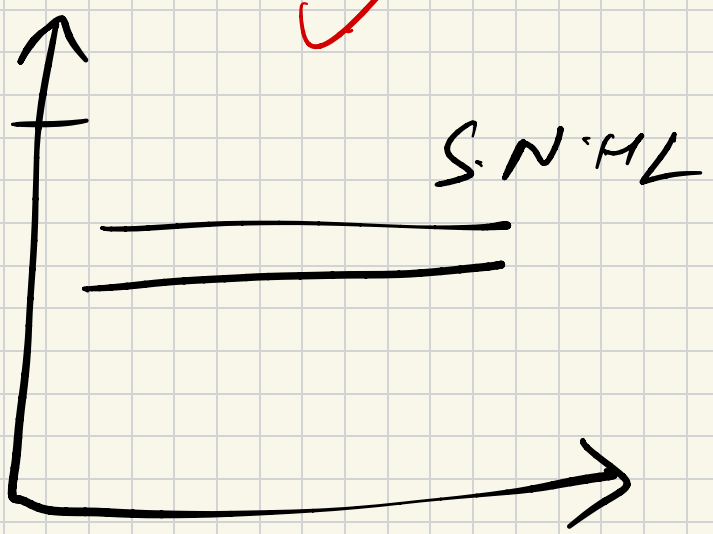






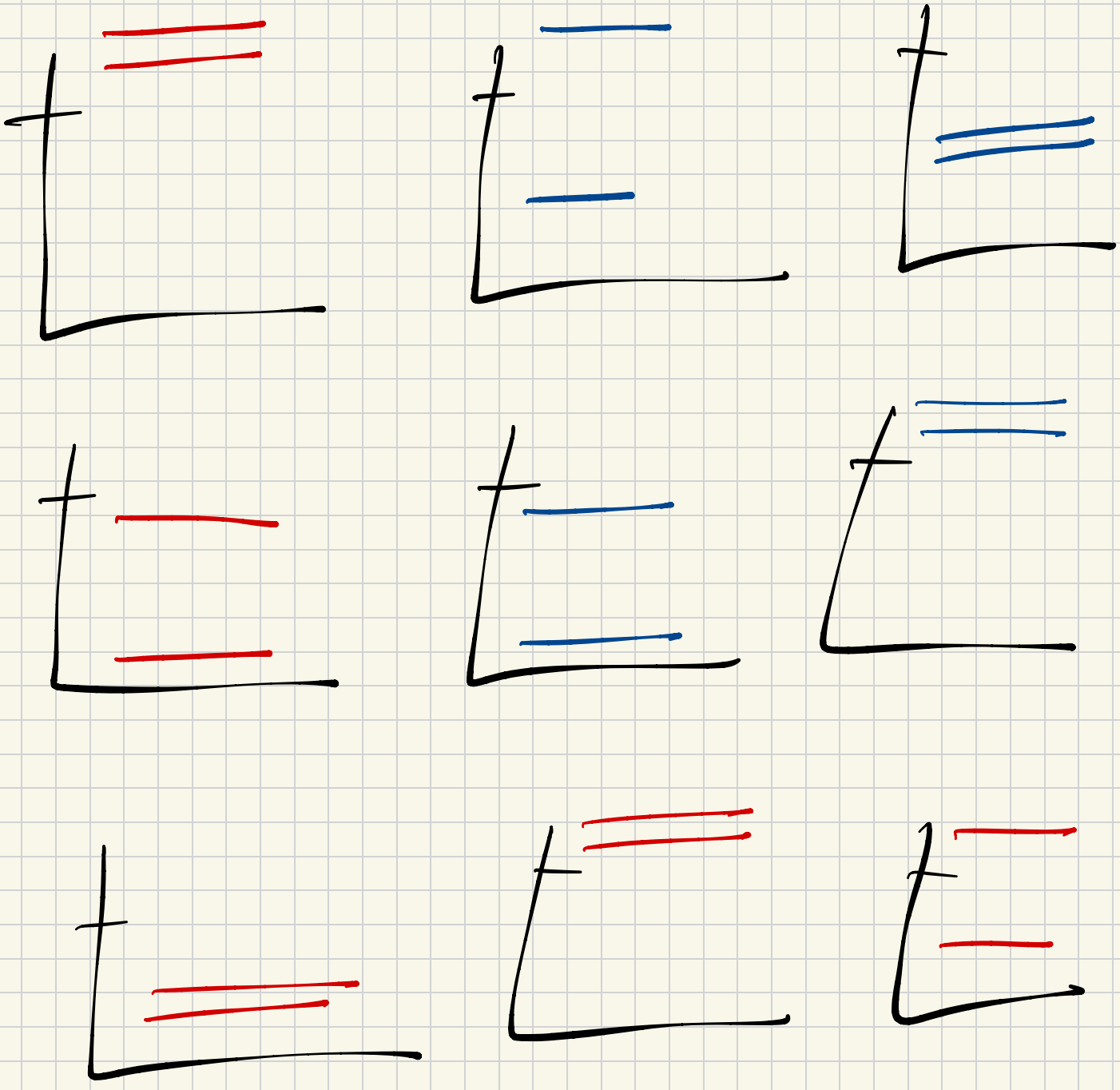
C.H.L

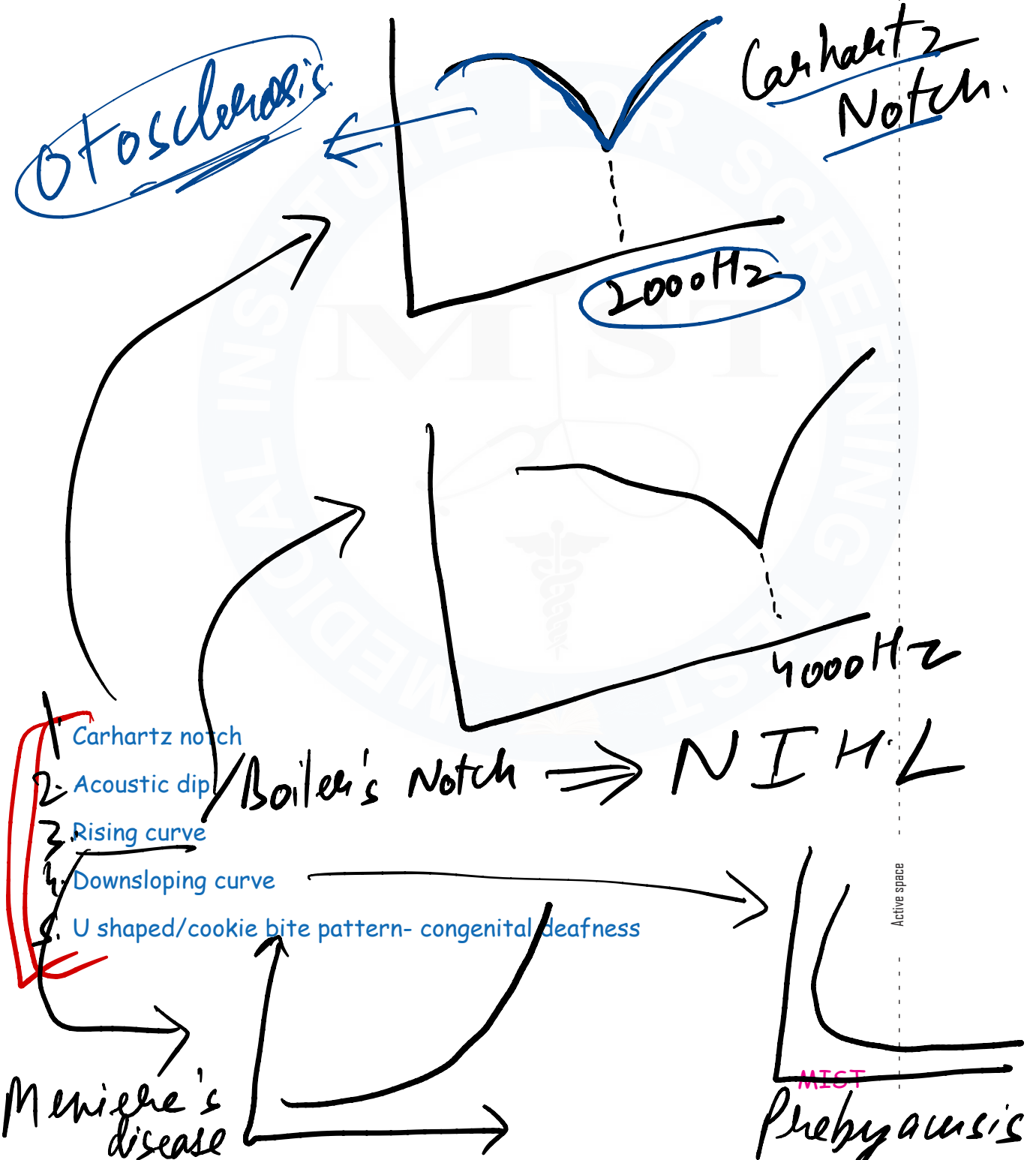




Red. → Right.

Blue → left.




Otosclerosis
Carhartz Notch
2000Hz
4000Hz
Boiler's Notch ⇒ NIHL
1. Carhartz notch
2. Acoustic dip
3. Rising curve
4. Downsloping curve
5. U shaped/cookie bite pattern- congenital deafness
Meniere's disease
Presbyacusis
Active space
MIST

OTOTOXIC DRUGS

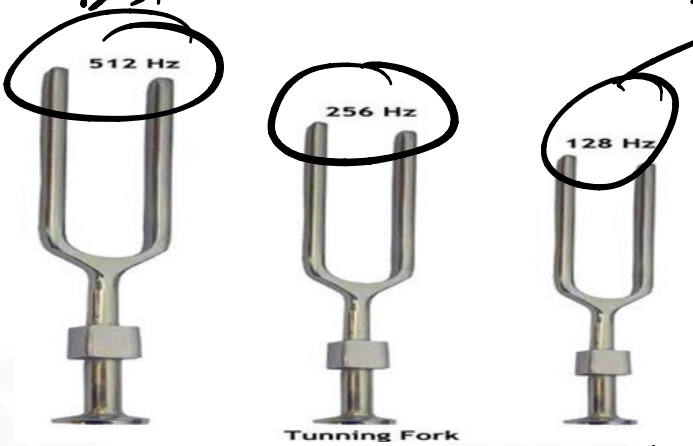
1. Aminoglycosides
2. Cisplatin
3. Diuretics

IN NEURAL DEAFNESS i.e RETROCOCHLEAR LESIONS

Nerve fatiguability test = tone decay test= threshold decay test= very poor sds score

Active space

TUNING FORK



Rinne's test

Weber test

Absolute bone conduction test

Bing test – alternative tragus press → snhl

Gelle test – siegelisation → otosclerosis

Activate Space

AVERAGE HEARING LOSS IN DIFFERENT LESIONS

Complete obstruction of ear canal → 30db

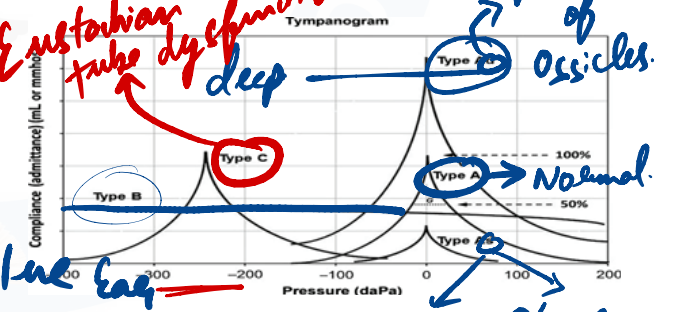
Perforation of tm (depends on size of perforation)-->10-40 db

Ossicular interruption with intact T.M → 54db

Ossicular interruption with perforation → 38 db

IMPEDENCE AUDIOMETRY

Tympanometry + Stapedial reflex



Active space

STAPEDIAL REFLEX

> 70 db SOUND GIVEN → STAPEDIUS MUSCLE CONTRACT TO DAMPEN SOUND OR TO PROTECT

ELECTROCOCHLEOGRAPHY

Electrical functions of Cochlea.

Action Potential

Summation Potential

SP/AP RATIO

S.P / A.P ratio < 0.3 N

< 0.3 → NORMAL

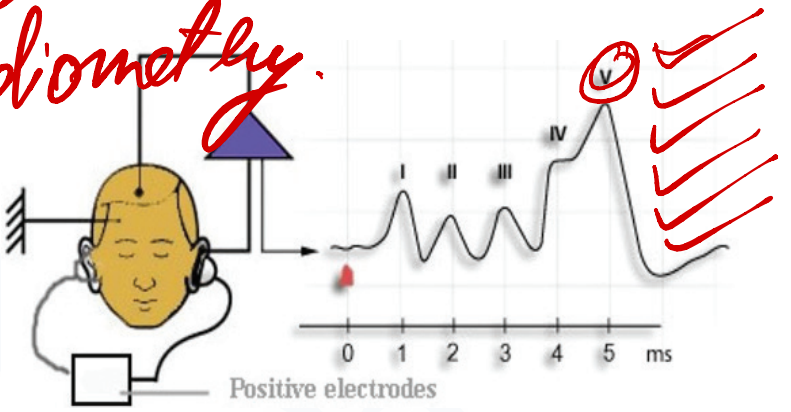
>0.45 → ABNORMAL

ECoh. IxOc → Meniere's disease

MIST

Active space

BERA → Brainstem Evoked Response Audiometry.



→ Best ear Ix.

→ Ix OC for Neural Hearing loss.

E- I, II

C-III

O-IV

L-V

E- N. VIII, Cochlear Nucleus

C- Olivary complex

O- Lateral Lemniscus

L- Inf. colliculus

I- MGB

A- Auditory Cortex

Medial geniculate body.

V is the largest wave

I is most sensitive wave

Active space

OTOSCLEROSIS

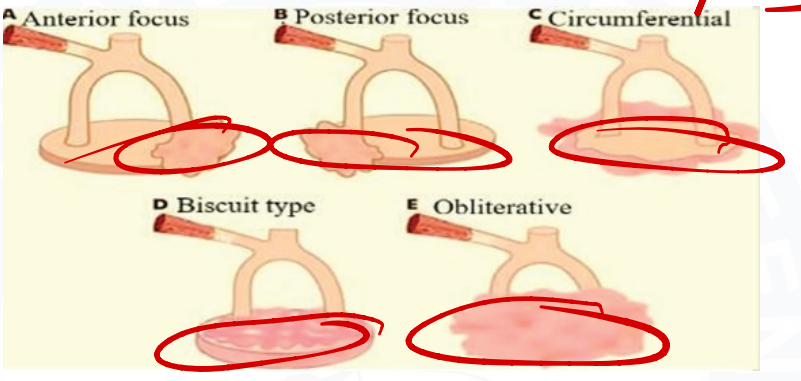
B/L

F>M

Most imp predisposing factor → hereditary

Associated with pregnancy

And measles



Vander hoeve syndrome

Conductive deafness- low intensity /mild type

Bilateral ,painless

In pregnant females commonly Paracusis willisi

Tinnitus and vertigo less common Speech will be soft

MISP

Schwartz / flamingo pink sign Carhartz notch

In active / otospongiotic phase sodium fluoride can be given

Stapedotomy -it is better → carhartz notch disappears after stapedotomy Stapedectomy

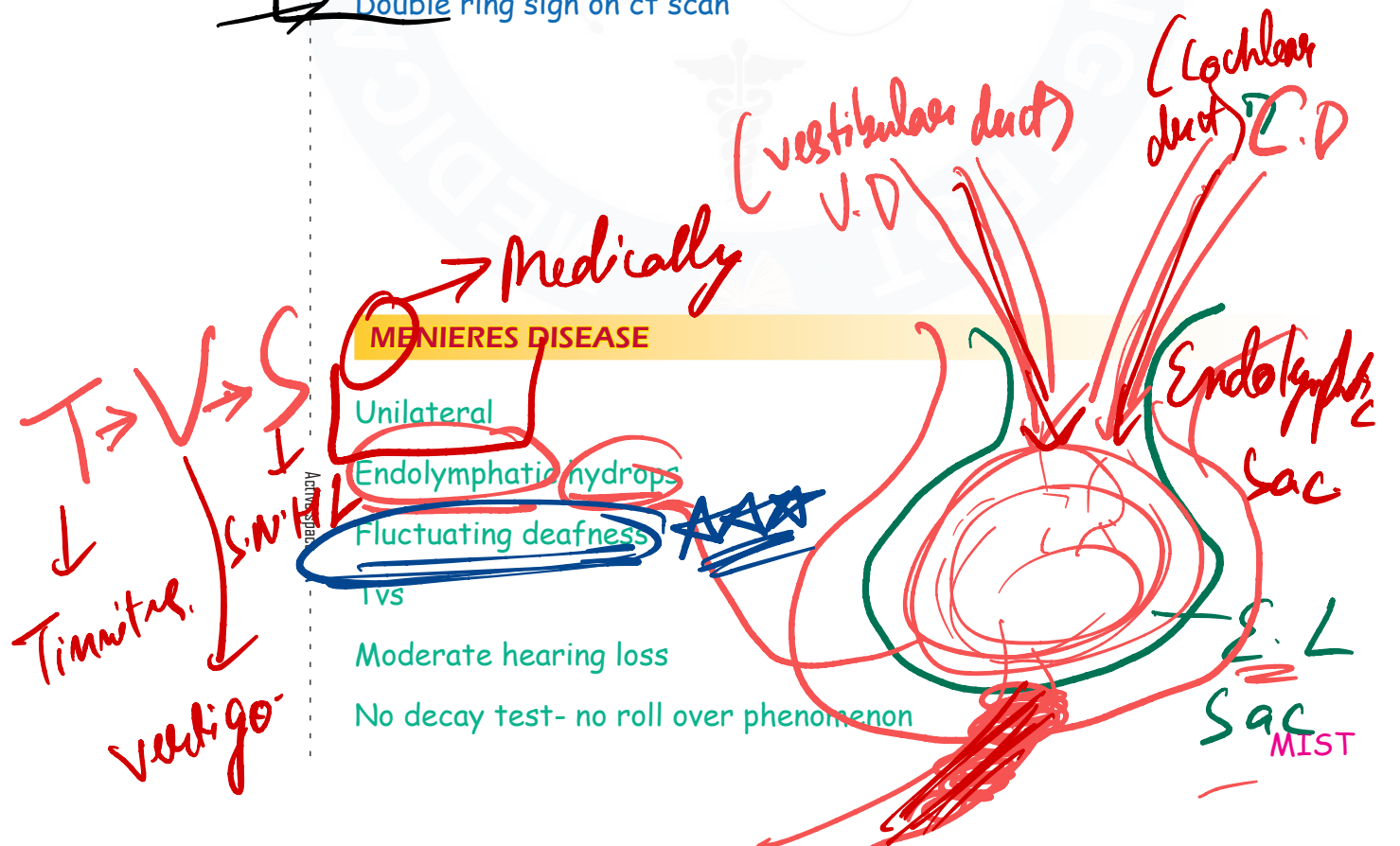
+

COCHLEAR OTOSCLEROSIS

Cookie bite audiogram

+

Double ring sign on ct scan



MENIERES DISEASE

Unilateral

Endolymphatic hydrops

Fluctuating deafness

Tvs

Moderate hearing loss

No decay test- no roll over phenomenon

MIST

Rinne- +ve

Weber lateralised to normal ear

T → V → S

Ioc- ecog

>0.45

SP

AP

Lermoyez/reverse menieres → S → V → T

Tulio phenomenon → Vertigo in loud sound.

Tumarkin crisis → H/o drop attacks

Diplacusis willisi- perception of 2 sounds as one ear is normal

Recruitment phenomenon - 1db change will be felt a big change

Glycerol test - give orally, its a dehydrating agent that reduces endolymph pressure → improves hearing

Now test.

Treatment

Medical

→ Vertin → β-histadine.

Vestibular suppresants medications

Low salt diet

Meniete device → Low pressure Intratympanic therapy.

MIST

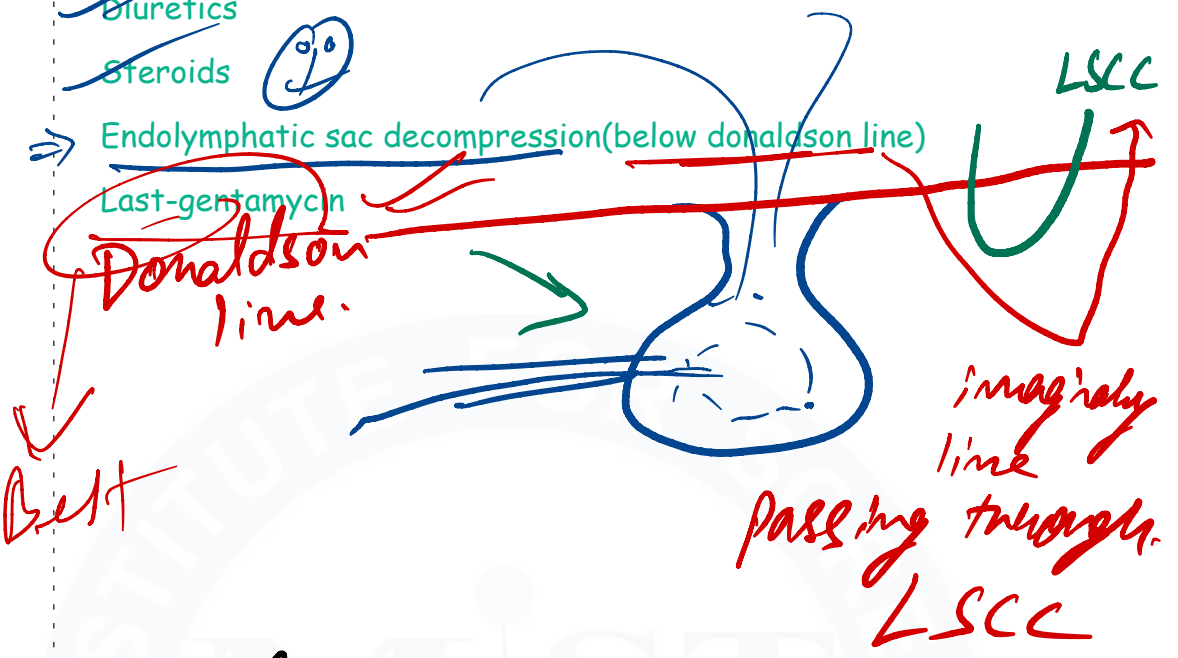
Active space

Diuretics

Steroids

⇒ Endolymphatic sac decompression(below donaldson line)

Last-gentamycin



Neural disease.

ACOUSTIC NEUROMA/VESTIBULAR SCHWANNOMA

M/c tumour of cerebellopontine angle

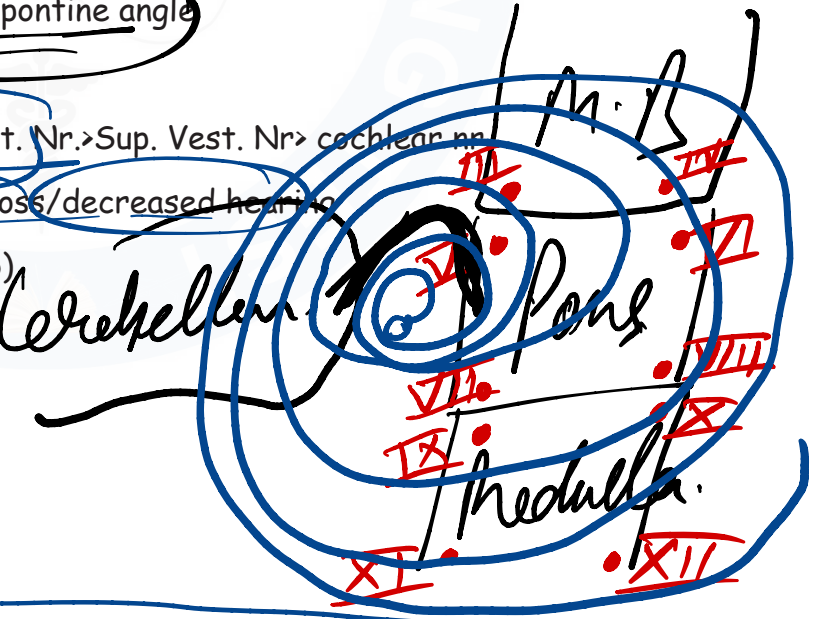
Nerve involved - inf. Vest. Nr.>Sup. Vest. Nr> cochlear nr

M/c symptom - hearing loss/decreased hearing

SEVERE deafness(>80db)

Roll over +

Decay test +



Ioc- bera

It causes pressure symptoms on all cranial nerves except i and ii cranial nr

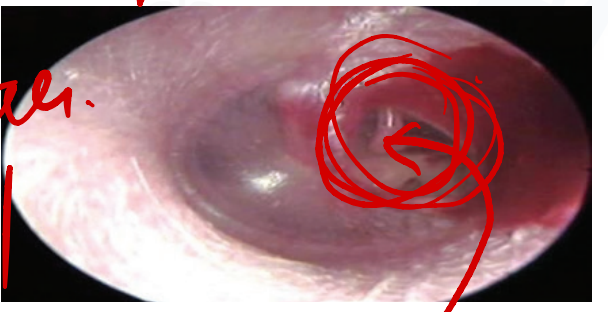
Active space

Radiological ioc- gadolinium enhanced mri

T/t - surgery

If patient denies surgery then gamma knife radiotherapy

TRAUMATIC EAR PERFORATION MANAGEMENT



Keep ear dry.

Active space

INFECTIONS OF EAR

Otitis externa → Infection of Ext. ear.

Otitis media → Infection of Middle ear.

Labyrinthitis → " " Inner ear.

Serous otitis media (glue ear) → infection occurs → asom Acute suppurative otitis media.

M/C OR THE EARLIEST FEATURE OF ET DYSFUNCTION IS RETRACTED TM

(-ve Middle ear pressure)

TOSS CLASSIFICATION PARS FLACCIDA

- **Grade I**: The pars flaccida is not in contact with the neck of the malleus.
- **Grade II**: The retracted pars flaccida is in contact with the neck of the malleus and clothing it.
- **Grade III**: Here in addition to grade II features there is minimal erosion of the outer attic wall
- **Grade IV**: In this grade in addition to all the above said changes there is severe erosion of the outer attic wall or scutum.

Active space

SADE CLASSIFICATION

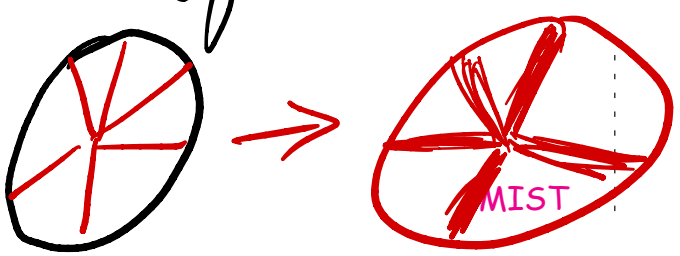
TABLE I.
Classification of Retraction Pockets of Pars Tensa as Proposed by Sadé.

Stage	Description
1	Slight retraction of tympanic membrane
2	Retraction of tympanic membrane, touching the incus or stapes
3	Tympanic membrane touching the promontory
4	Tympanic membrane adherent to the promontory

SEROUS OTITIS MEDIA(GLUE EAR) AND ASOM

Stages of asom → 1. Stage of Tubal occlusion
2. Stage of Pre suppuration.
3. Stage of Suppuration.
Grommet → 4. Stage of Resolution.

Signs of ASOM ⇒ 1. Cart wheel appearance of T. Membrane.



TLT ⇒ 1. Antibiotics.

2. Analgesics.

3. Nasal decongestants

~~*****~~ ** 4. Myringotomy + Grommet insertion

II

Anterioinferior

Eardrum

↳ "Beer can opening principle"

COMPLETE CSOM - SAFE AND UNSAFE

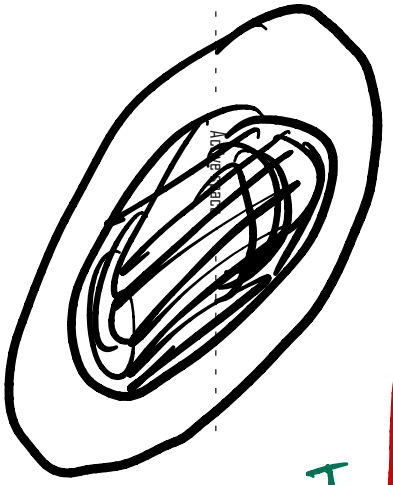
RESERVOIR SIGN

Safe / Tubotympanic COM → mucosal | Unsafe / Atticoantral / COM-Squamous.

WULLSTEIN CLASSIFICATION OF TYMPANOPLASTY

TYPE

- I - Myringoplasty
- II - Myringoincudopexy
- III - Myringostapedopexy
- IV - On foot plate(minor/small cavum effect)
- V - Fenestration surgery



MISTIOSFORM

I II III IV V

MIST

OSSICULAR CHAIN RECONSTRUCTION

Porp → Partial ossicular reconstruction prosthesis

Torp → Total ossicular reconstruction prosthesis

T.M.

MATERIALS USED FOR [redacted] RECONSTRUCTION

Temporalis fascia (M/C)

Perichondrium

Periosteum

Artificial graft (alloderm)

APPROACHES FOR THE SAME

Rosen

Transcanal approach is the most commonly used approach

Except in pediatric patients where endaural and postaural approach are used

Active space

CHOLEASTETOMA

Metaplasia of the eroded epithelium

Prussack space- m/c and earliest site

Sinus tympani- residual and reoccurence

DX OF CHOLESTEATOMA

It generally presents as white mass behind t.M

It can be:-

- ✦ Paediatric- congenital cholesteatoma
- ✦ Retracted t.M w/o infection- primary cholesteatoma
- ✦ H/o infection, t.M perforation - sec. Cholesteatoma
- ✦ H/o any trauma or surgical trauma- tertiary cholesteatoma

HYPOTHESIS

Most accepted.

Retraction pocket formation- whitman's theory

A - 1

Basal cell hyperplasia- ruedis theory

B - 3

Squamous metaplasia- sade theory

Epithelial migration- haberman theory

MASTOIDECTOMY TYPES

Prem Kotwal

COMPLICATIONS OF CSOM

Reservoir Sign.

1. Mastoiditis

M/c overall or m/c extracranial- mastoiditis M/c symptom of mastoiditis- ear discharge M/c sign of mastoiditis- tenderness

Earliest sign - fraying of mastoid → Mastoid becomes smooth & shiny

Hallmark sign- sagging of canal → Postwall

MIST

Mastoid abscess a/k/a Post auricular.

Active space

2. Abscess

Bezold → Pus in Sternocleido mastoid.

Citelli → Pus in Post belly of digastric.

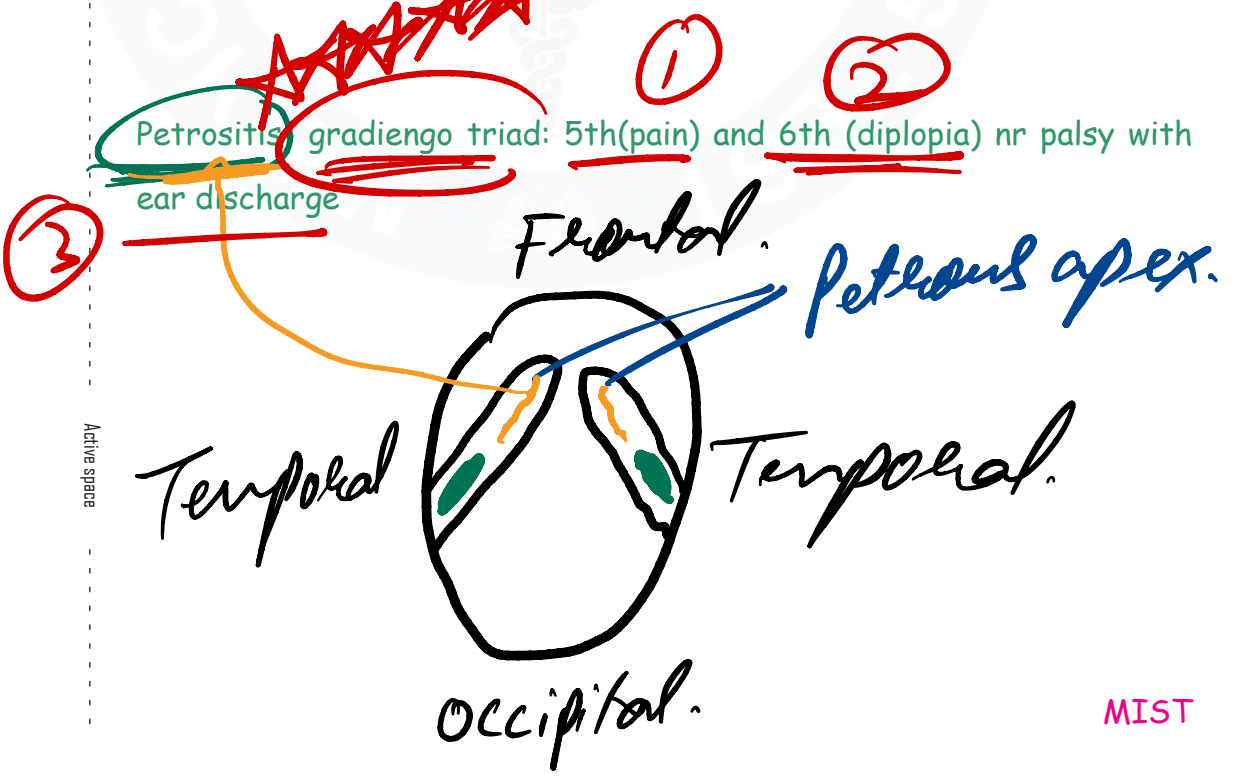
Luc abscess → Pus in deep bony part of E.A.C

M,c is post auricular abscess

3. Facial Nerve dysfunction → M/c part of the facial nr. Involved is tympanic → Horizontal segment

4. Labyrinthitis

5. Petrositis, gradiengo triad: 5th(pain) and 6th (diplopia) nr palsy with ear discharge



Active space

REMEMBER ALSO

Mac ewans triangle boundaries

Sinodural angle also called as citelli angle

INTRACRANIAL COMPLICATIONS

M/c meningitis

2nd m/c brain abscess

Subdural abscess

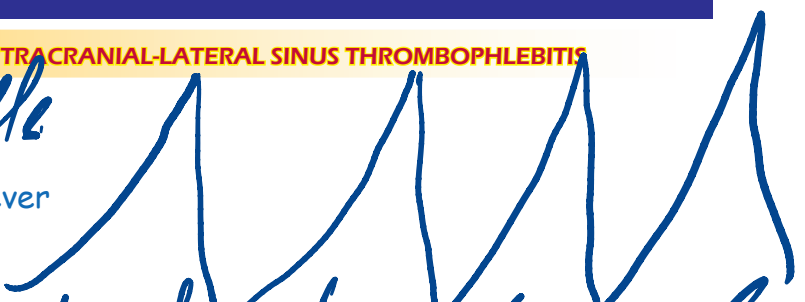
Otic hydrocephalus

Active space

LAST IMP COMPLICATION INTRACRANIAL-LATERAL SINUS THROMBOPHLEBITIS

Fever rises & falls

1. Hectic/picket fence fever



Battle Sign: 2. Gresinger sign — Bluish discoloration of Mastoid due to engorgement of ...

3. Crowe beck sign — Retinal Blood vessels engorgement leading to Papilloedema

4. Tobey ayer sign

↳ No change in C.S.F pressure despite pressing Jugular vein.

5. Delta sign on ct

Active space

MIST

INFECTIONS OF EXTERNAL EAR

BOXER EAR/ CAULIFLOWER EAR

Perichondritis of Pinna.

Active space

ALL ENT INFECTION CAUSED BY STREPTOCOCCUS F/B STAPH EXCEPT FURUNCLE CAUSED BY STAPHYLOCOCCUS MAINLY

Few caused by pseudomonas are

1. Perichondritis
2. Diffuse otitis externa
3. Malignant otitis externa
4. Csom

DIFFUSE OTITIS EXTERNA NAMES

Generalised otitis externa → Folliculitis in E.A.C

Swimmers ear

Singapore ear

Tropical ear

Telephonic ear

T/t ⇒ Antibiotics

Active space

MALIGNANT OTITIS EXTERNA/ OSEOMYELITIS OF SKULL BASE

Diabetes patient

org → Pseudomonas aeruginosa.

Presents as ear discharge + granulation in canal + can have facial palsy

Tc 99 scan for diagnosis → Bone scan.

Gallium scan for prognosis → checks inflammation.

T/t → Anti pseudomonal drugs

- Scott-Brown's Otorhinolaryngology Head n Neck Surgery staging (by combining three staging system published between 1985-1991)

Stage 1: clinical evidence of malignant otitis externa with soft tissue infection beyond external auditory meatus, but negative 99mTc bone scan

Stage 2: soft tissue infection beyond external auditory meatus with positive 99mTc bone scan

Stage 3: as stage 2 but with cranial neuropathy
- 3a: single
- 3b: multiple

Stage 4: as stage 2/3 with intracranial complications (meningitis, empyema, sinus thrombosis, brain abscess)

Death

Cefepime
Cef Perazone
Ceftazidime
Cefta Zidime
Piptaz

MIST

Active space

REMEMBER THESE DISEASES IN ENT FOR DIABETIC PATIENT

Ear- malignant otitis externa

Nose- mucormycosis (invasisve disease, liposomal amphotericin b for treatment)

Oral cavity- ludwig abscess

Neck

TUBERCULAR OTITIS MEDIA

Pale

Perforations(multiple)

Painless

T/T - ATT

P → Pale granulation.
P → Profuse discharge.
P → Perforations (multiple)
P → Painless.

Active space

OTOMYCOSIS

Wet newspaper like appearance

Aspergillus niger

OSTEOMA AND EXOSTEOSIS DIFFERENCE

Exosteosis aka surfer ear (fast moving water)

Active space

Kilkari Logo

MIST

VIRAL INFECTIONS OF EXT. AND MIDDLE EAR

Shingles in ear.

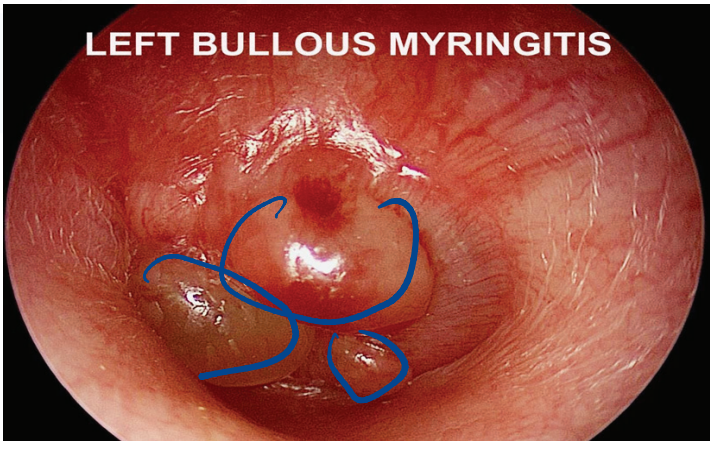
Herpes zoster oticus —> in immunocompromised patients

+ Facial Nerve palsy

RAMSAY HUNT SYNDROME

MYRINGITIS BULLOSA- PAINFUL BLEBS ON TM

infection / inflammation of T.M.



Cause:

Streptococcus pneumoniae.

Measles.

Active space

NEWBORN SCREENING FOR HEARING

Screening procedures of choice

Oae- after 24 hours/ 1st day should be done

Should be definitely done by 1st month

Bera - by 3rd month

But remember

FOR A NEWBORN IN ICU i.e high risk baby - Bera is the screening procedure of choice directly

HEARING AID

Components

R- reciever

A- amplifier

M- microphone

Active space

Use 1. Bte- behind the ear

2. Cic - completely in th ear

3. Ite- in the ear

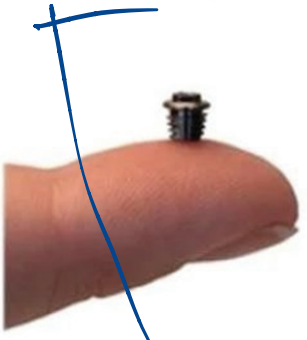
C/I for conventional hearing aid.

If canal atresia, canal stenosis, pus in the canal present then baha is used

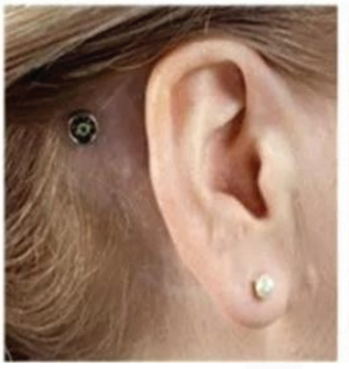
Active space

BAHA ⇒ Bone anchored Hearing Aid.

1. Titanium implant



2. Abutment



3. Sound Processor

BAHA

MEDICAL ENGINEERING TEST

Active space

MIDDLE EAR IMPLANT

External component is audio processor

Internal component is vorp(vibrating ossicular prosthesis)

COCHLEAR IMPLANT

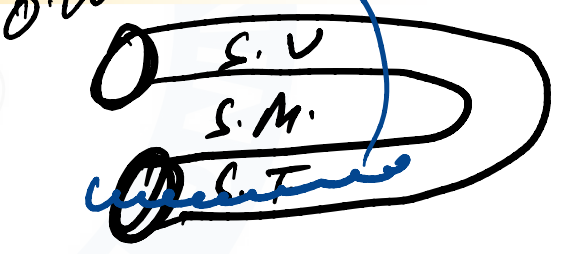
Mondini dysplasia is m/c indication

Michael aplasia - contraindicated in it

Facial recess approach.

Post lingual deafness are best candidates

Can be used in pre lingual deafness also



Active space

IN NEURAL DEAFNESS - BRAINSTEM IMPLANT IS USED - KEPT IN LATERAL RECESS OF FOURTH VENTRICLE

Fistula test

Press the tragus alternatively to check vestibular system(nystagmus)

-VE in normal people

+VE in cholesteatoma

False +ve means symptoms +sent in absence of fistula seen in menieres disease and congenital syphillis-hennebert sign

False -ve means symptoms -sent in presence of fistula seen in dead labyrinth and [redacted]

GLOMUS TUMOUR/PARAGANGLIOMA

Glomus jugulare and glomus tympanicum

Signs:

Rising sun sign - tumour coming from below

Brown sign - blanching on pressing t.M (by siegelissation)

Aquino sign - blanching on pressing neck vessel by (siegelisation)

Carotid and jugular bony partition lost - phelps sign

Active space

SUDDEN SNHL

30db

3 consequtive frequencies

3 days

Causes

M/c is vascular & idiopathic

T/t is generally steroids

Active space

CALORIK AND HALLPIKE TEST

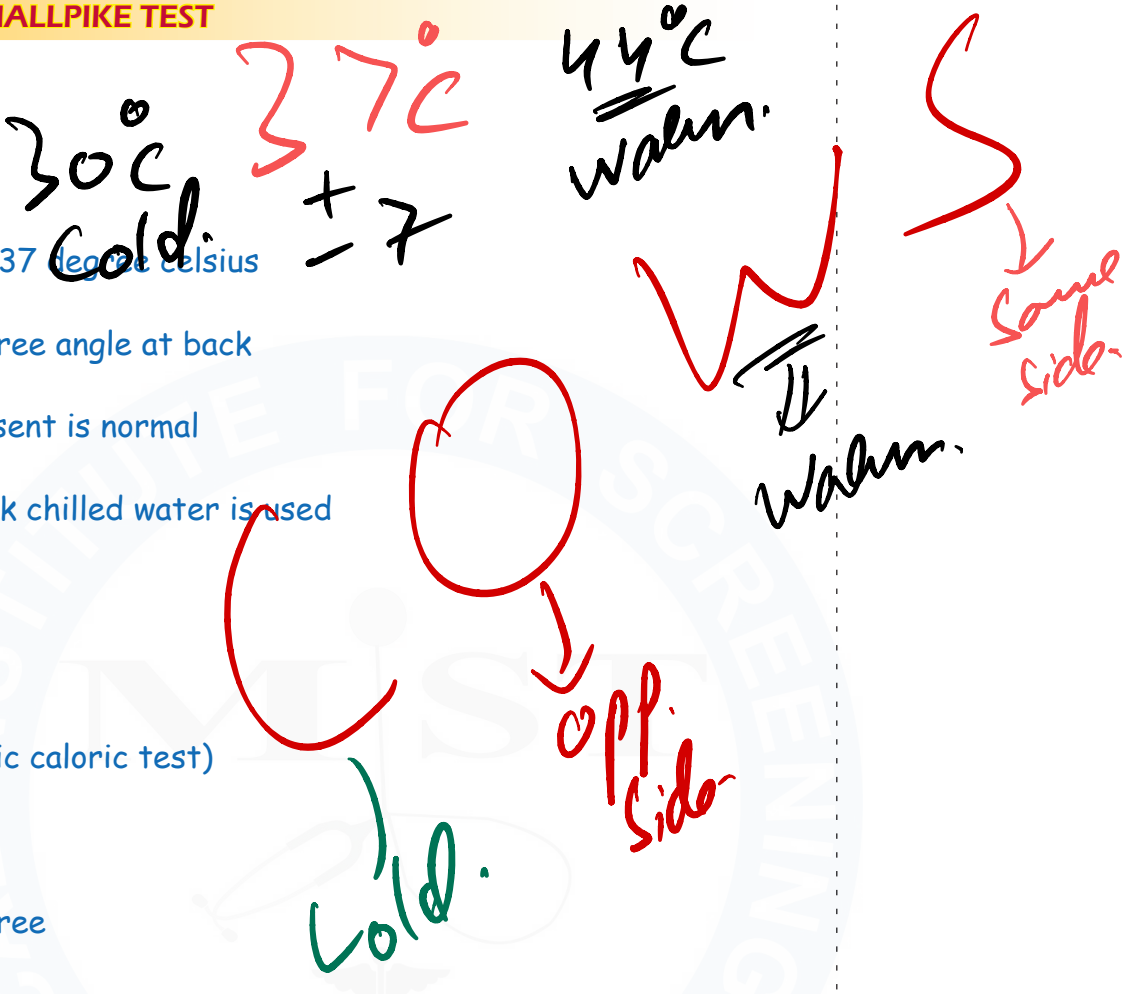
Caloric test

Temp of water is 37 degree celsius

Position is 60 degree angle at back

If nystagmus present is normal

In modified kobrak chilled water is used



Hallpike (bithermic caloric test)

Cows

Position is 30 degree

Active space

VEMP → Vestibular Evoked Myogenic Potential.

- cVEMP- Scm muscle inhibitory response-> inf vestibular nr.
  - Cervical.
- oVEMP- Extraocular muscle excitatory response-> sup. Vestibular nr.
  - Ocular.

Active space

Every time with sudden changes in head position person has

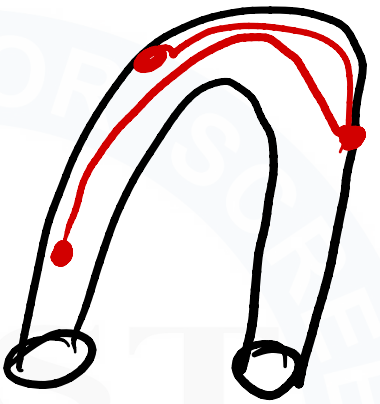
EAR

BPPV → Benign Paroxysmal Positional Vertigo

Due to otolith / otoconia made of calcium carbonate

Mostly post. Scc

Diagnosis by - dix hallpike manuevere



→ Vertigo

Treatment by

Epleys manuvere 2

Semonts manuevere 3

Brandt and daroff manuvere 4

Active space

LARYNX

Consists of three paired and three unpaired cartilages

Unpaired	Paired
Thyroid	cuneiform
Cricoid	corniculate
Epiglottis	arytenoids

Thyroid angle - angle between two thyroid lamina

90° → Males. (Adam's apple)

120° → Females.

Active space

Larynx Is Lined By Ciliated Columnar Epithelium Except Vocal Cords That Are Lined By Stratified Squamous Epithelium

Laryngeal Crepitus is Normal and if absent Δ → Ca Larynx. → "Bocca's sign"

Three divsions of larynx

1. Supraglottis- epiglottis, ae folds, false vc, ventricles, saccule → arytenoid. → ant-most part of ventricle space

II Space Blw True & False V.C

2. Glottis - true vocal cord length of vocal cord

3. Subglottis

Active space

Length of vocal cord

MALE- 18-23 mm

FEMALE-16-17 mm

Males have low pitch(dull)

voice Females have high pitch(sharp) voice

DISORDERS

1. Puberphonia- feminine voice in adult male

2. Androphonia- masculine voice in female

⇒ T/t

⇒ Thyroplasty

Active space

REINKES EDEMA → collection of fluid in subepithelial space of V. cord.

Cause

→ Smoking

→ Voice abuse

Dx → Endoscopy

T/t → MLS

Microlaryngeal Sx.

Active space

VOCAL NODULE/SCREAMER NODULE/TEACHER NODULE

Cause- voice abuse

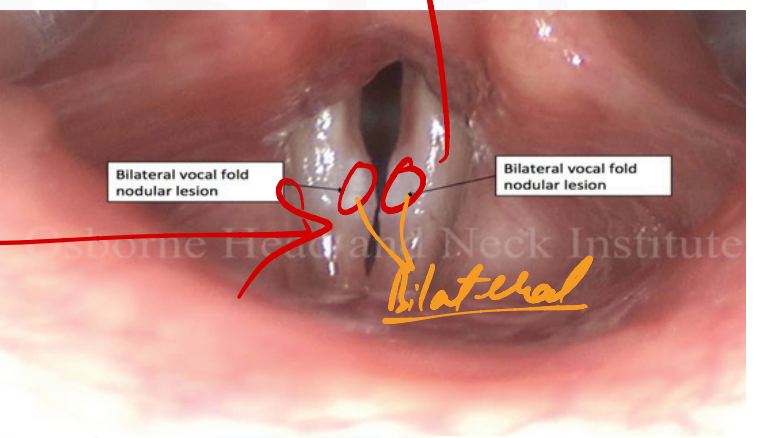
Lpr

Always bilateral

Non Smoker.

No Surgery.

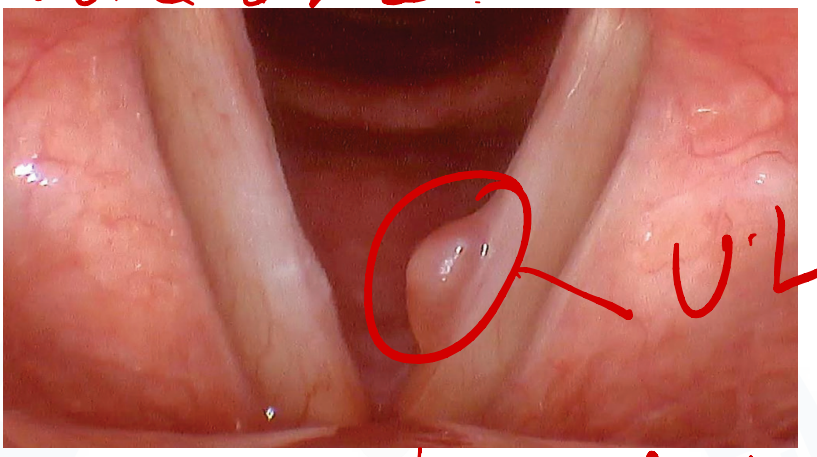
Rx → P.P.I + Voice Rest.



Active space

VOCAL POLYP

↳ Smokers
↳ Voice abuse.



T/t → M.L.S
Micro laryngeal Sx.

Pediatric Vs. Adult

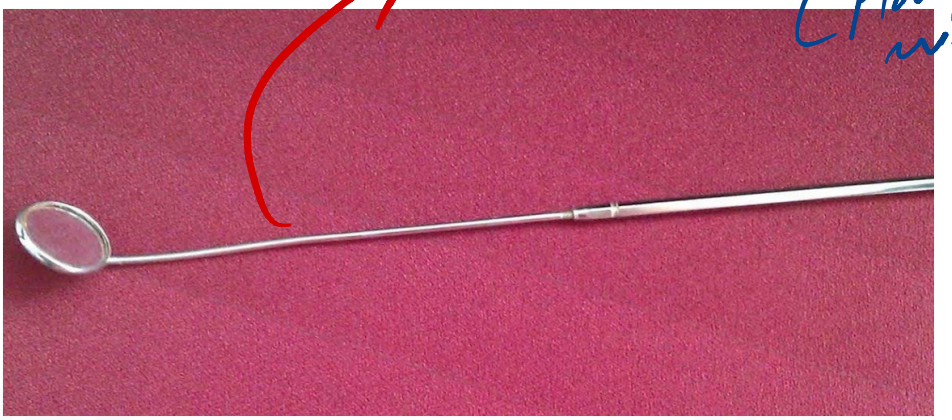
Significant Anatomical Airway Differences between a Pediatric Patient and an Adult Patient

Airway	Pediatric	Adult
Tongue	Large relative to oropharynx	Normal
Epiglottis shape	Floppy, short and omega shaped	Firm, flatter
Epiglottis level	Larynx higher at C3-C4	C5-C6 Lower.
Larynx shape	Funnel shaped	Columnar/cylindrical
Narrowest portion	At cricoid cartilage (subglottic)	At vocal cords Glottis.

Active space

MIST

METHOD OF EXAMINATION OF LARYNX



IDL DOESN'T SHOW

- Laryngeal surface of epiglottis
- Ant.Commisure
- Ventricles
- Undersurface of vocal cords

Active space

FUNCTIONS OF LARYNX

Protection of airway

Sound production

Stridor- noisy breathing due to airway obstruction

Inspiratory-supraglottis

Expiratory- trachea and lungs

Biphasic-glottis, subglottis

Active space

MUSCLES OF LARYNX

Adductors → Protection of Airway → Sound production.

- Cricothyroid(main tensor along with vocalis muscle)
- Thyroarytenoid
- Lateral cricoarytenoid
- Interarytenoid

Abductor → Breathing / Respiration.

- Pca(safety muscle of larynx)

Post Cric Arytenoid.

Vagus

Motor supply nerve of larynx

All intrinsic are supplied by rln except cricothyroid which is supplied by ext branch of sln

P.C.A

Superior laryngeal Nerve

- External
- Internal

Recurrent laryngeal Nerve

MIST

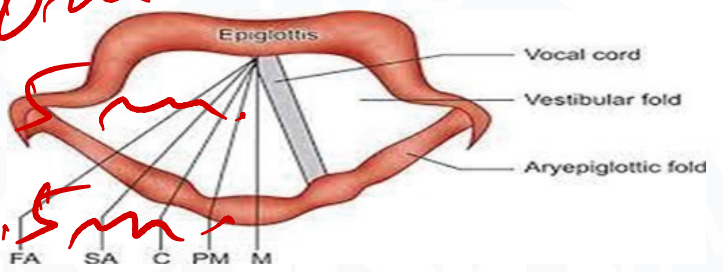
Active space

Sensory supply of larynx

Supraglottis - internal br. Of sln

Glottis - mainly rlnX also internal br. Of sln Subglottis-rln

0,1.5,3,7,9 mm Positions Of VC



Median → 0 mm

Paramedian → 1.5 mm

Cadaveric → 3.5 mm

Partial Abduction → 7 mm

Full Abduction → 9 mm

ONE LINERS

Incidence of injury rln >> sln (RLN has a longer course)

Left rln >> right rin (left has longer course)

In thyroid surgery- ext. Branch of sln

M/c/c of u/l vocal cord palsy- malignancy(bronchus ca)

M/c/c of b/l vocal cord palsy- thyroid surgery

Active space

Tracheostomy

→ M/C indication → Prolonged intubation.

→ M/C type of tube → Portex cuffed Tracheostomy tube.

Types

- High. 1st & 2nd T. ring. → Laryngeal
- Mid 2nd & 3rd T. ring. → M/C
- Lower 3rd & 4th. → FB aspiration

VOCAL CORD PALSY

U/L RLN (ABDUCTOR) PALSY → No emergency

B/L RLN (ABDUCTOR) PALSY → Tracheostomy + Type II thyroplasty

U/L SLN (ADDUCTOR) PALSY → No emergency

B/L SLN (ADDDUCTOR) PALSY → Tracheostomy + Type I thyroplasty

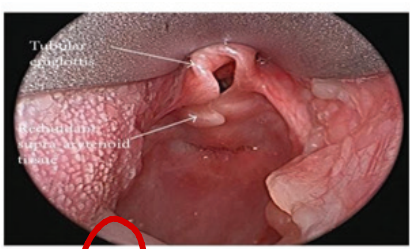
LARYNGOMALACIA

Laryngeal cartilages weak, flaccid

⇒ M/c congenital anomaly of larynx

M/c c/c of congenital stridor.

Stridor decreases on prone position



inspiratory stridor.

Δs → Endoscopy → Ω omega shaped epiglottis

MIST

Active space

Isshiki

Thyroplasty

Alteration in length & position of V. cord.

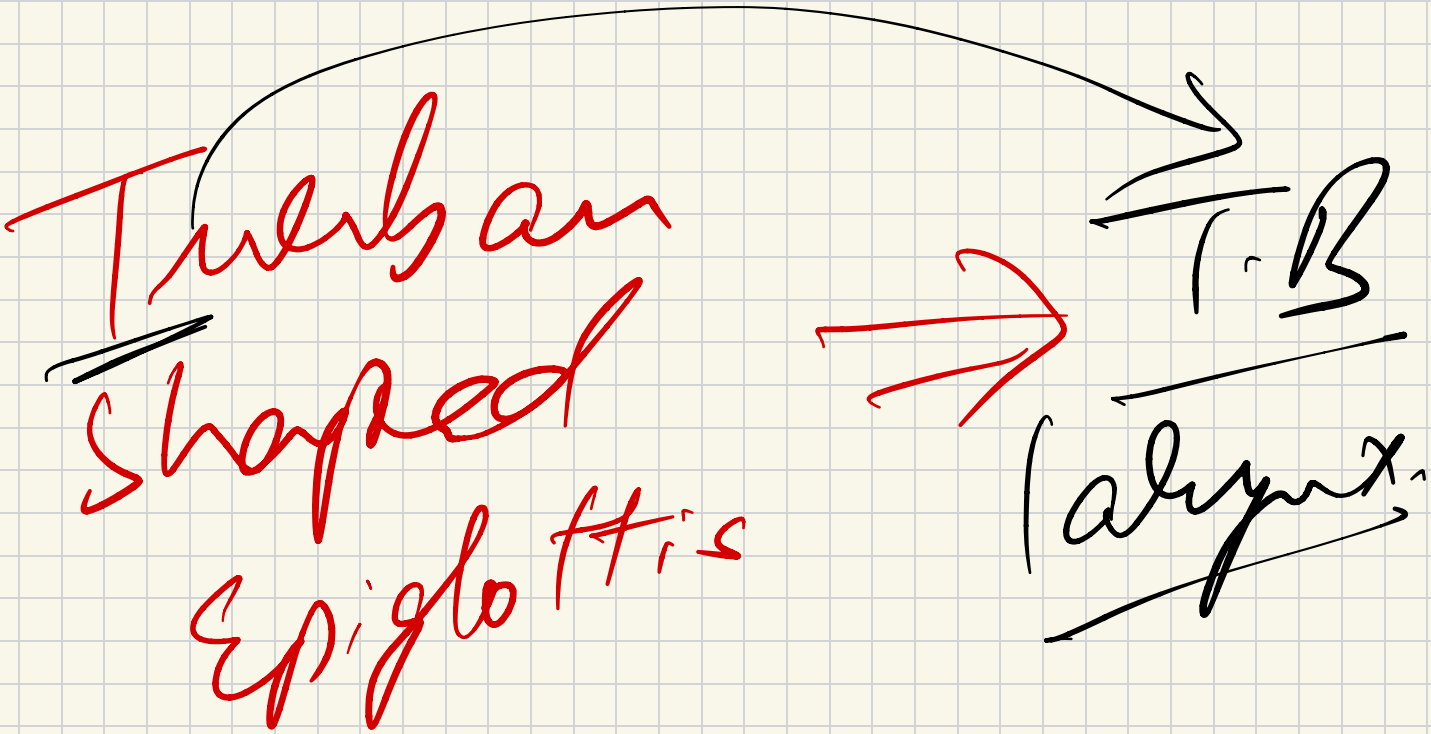
I M → Medialisation → S.L.N palsy.

II L → Lateralisation → R.L.N palsy.

III S → Shortening → Puberphonia.

IV Lengthening → Androphonia.

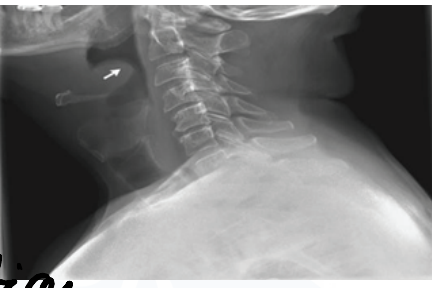
T/t ⇒ Reassurance



EPIGLOTTITIS

Cause: H. influenza > S. pneumoniae

Child has voice change, sometimes Resp. distress



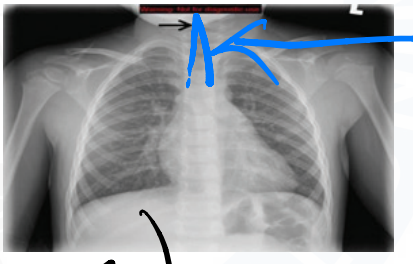
Tripod position.

X-Ray sign → Thumb sign.

T/t → Antibiotics + O2 Humidifiers

CROUP- STEEPLE SIGN

L.T.B → Laryngotracheobronchitis



Cause → Parainfluenza virus (Type B > Type A)

X-Ray Sign → STEEPLE sign.

T/t → Steroids + Humidified O2.

MIST

Active space

SOME POINTS TO DISCUSS AT LAST

Tonsills and tonsillectomy

Adenoids and st clair curette

Styalgia/ eagle syndrome

Ludwig angina

Tracheostomy

Active space

Workbook Question

1. Carhart's notch is characteristically seeh at
   a) 1000hz
   b) 2000hz
   c) 4000hz
   d) 6000hz

2. Gradenigo's syndrome does not consist of
   a) abducent nerve palsy
   b) Retro orbital pain
   c) aural discharge
   d) Palatal palsy

3. Endolymphatic hydrops is associated with
   a) Otosclerosis
   b) CSOM
   c) wax impacted ear
   d) Menieres disease

4. Positive Rinne test is seen in
   a) Otosclerosis
   b) csom
   c) wax impacted ear
   d) Presbycusis

Active space

5. Hyposthesia of the posterior aspect of the external auditory canal may be an early sign of
   a) Trigeminal neuralgia
   b) Menieres disease
   c) Lateral sinus thrombosis
   d) Acoustic neuroma

6. Pulsatile tinnitus in ear is due to
   a) Malignant otitis media
   b) osteoma
   c) mastoid reservoir
   d) Glomus tumour

7. A patient hears better in Noise the diagnosis is
   a) Hyperacusis
   b) Hypoacusis
   c) Presbycusis
   d) Paracusis

8. Tone decay test is done to find out
   a) Otosclerosis
   b) Cochlear lesions
   c) Retrocochlear lesions
   d) Menieres disease

Active space

9. Blue drum is seen in

A) Tympanosclerosis

B) Secretory Otitis Media

C) Otosclerosis

D) Myringitis Bullosa

10. A 30 year old male is having attic cholestaetoma of left ear with lateral sinus thrombophlebitis. Which of the following will be operation of choice?

A) Intact canal wall mastoidectomy

B) Simple mastoidectomy

C) Canal wall down mastoidectomy

D) Mastoidectomy with cavity obliteration

11. The normal length of external auditory meatus is

a) 7mm

b) 10mm

c) 24mm

d) 36mm

12. Mc Evans triangle is the landmark for:

a) Maxillary sinus

b) Mastoid antrum

c) Frontal sinus

D) None

Active space

13. Multiple perforations of Tympanic membrane most likely will be seen in

a) TB otitis media

b) ASOM

c) CSOM

d) Glomus tumour

14. A pregnant woman in 3rd trimester complains of hearing her own voice.Examination of ear shows movement of Tympanic membrane synchronus with her respiration which is especially exagerrated after nostril occlusion of contralateral side . Most probable diagnosis is ?

A) Otosclerosis

B) Patulous eustachian tube

C) Eustachian tube obstruction

D) Otitis media with effusion

15. Treatment of Dry traumatic rupture is :

A) Antibiotic ear drops

B) Myringoplasty

C) Ear pack soaked with antibiotic

D) Protection of ear against water

16. Hyperacusis in Bells palsy is due to paralysis of the following muscle:

A) Tensor tympani

B) Tensor veli palatini

C) Stapedius

D) Levator veli pelitini

Active space

Workbook Sheet

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

Workbook Sheet

Active space

Workbook Sheet

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

Workbook Sheet

Active space

Workbook Sheet

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

Workbook Sheet

Active space

Workbook Sheet

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

Workbook Sheet

Active space

Workbook Sheet

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

Workbook Sheet

Active space

Workbook Sheet

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

Workbook Sheet

Active space

Workbook Sheet

Active space

Workbook Sheet

Active space

Workbook Sheet

Active space